



AVEVA™ Predictive Analytics Web

API Guide

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Chapter 1

Understanding the AVEVA Predictive Analytics Web API

AVEVA Predictive Analytics Web REST API overview

The AVEVA Predictive Analytics Web REST API (REST API) exposes high-level information about your assets, projects, and alert states. Use the REST API to:

- Retrieve historical data from your system.
- Display the data in your custom reporting, dashboards, case library, and other AVEVA Predictive Analytics Web features.
- Manage alert states.

RESTful API

All methods are exposed as RESTful functions. These return JSON results and use standard HTTP return codes and HTTP verbs, such as:

- GET
- POST
- UPDATE
- DELETE

Token-based security

Secure authentication uses tokens for its transactions. These identify the user and are included in the authorization header.

Strongly versioned

To ensure backwards compatibility, API calls are strongly versioned. As methods change over time, there is no urgency to update API calls.

API functional areas

Use these functions to create your own applications to leverage your AVEVA Predictive Analytics Web interface and your historical data.

- **Alert Status**

The functions return the alert status (including security limited roll-ups) for an asset and include results for the child assets and projects and points.

- **Point Attributes And History**

These functions return historical data for one or more tags in a project.

Available alert functions are:

- **Alert Management Lists**

Two sets of queries provide information to populate the dashboard, including:

- Alert states
- User-defined values available when the "clear alert" option is provided

- **Alert Management States**

Endpoints set the alert workflow state for a resource (asset, project, or point). The request body also includes:

- Expiration time
- User-provided notes
- Other details

Chapter 2

Use the API

General organization

API calls are based on the API root. For example, if AVEVA Predictive Analytics Web is installed in the *AVEVAPredictiveAnalytics* directory, then the root for the API is **avevapredictiveanalytics/api/**.

The general URL format is:

/endpoint/datatype/record?filter=xxx&sort=xxxx&page=xxxxxx&other_parameter=xxxxxx

For example, **api/projects/123** provides information about project 123.

Some endpoints may include optional parameters. See the endpoint definitions for more details.

Although the 2019R1 version of the API works seamlessly with legacy calls, before upgrading to the 2019 R1 API, it is recommended that you explicitly use V1 in the URL. For example:

`https://localhost/api/v1/assets/100`

Supported calls

The AVEVA Predictive Analytics Web REST API exposes key parts of the AVEVA Predictive Analytics system through a series of REST endpoints.

The following calls are supported:

- /api/identity
- /token
- /api/assets/ or api/assets/[assetID]
- /api/projects/{ProjectID}
- /api/points/{PointID}
- /api/histdata/
- /api/AlarmWorkFlowStates
- /api/AlarmWorkFlowStates/ClearParameters
- /api/assets/{id}/AlarmWorkFlowState
- /api/projects/{id}/AlarmWorkFlowState
- /api/points/{id}/AlarmWorkFlowState

- /api/AlertThresholds/new-point-thresholds
- /api/AlertThresholds/thresholds (PUT)
- /api/AlertThresholds/thresholds (DELETE)
- /api/AlertThresholds/template-references-reset
- /api/AlertThresholds/thresholds/{ThresholdId}
- /api/AlertThresholds/point-thresholds/{pointId}
- /api/CalculationPoints (GET)
- /api/CalculationPoints (PUT)
- /api/DigitalGroups
- /api/FaultDiagnostic
- /api/FaultDiagnostic/project/{projectId}/{frequencySeconds}
- /api/FaultDiagnostic/project/{projectId}/{frequencySeconds}/recentMatchPerentage
- /api/FaultDiagnostic/fault-details
- /api/FaultDiagnostic/summary
- /api/Forecast
- /api/HistorianPoints (GET)
- /api/HistorianPoints (PUT)
- /api/HistoricalArchive/omr/{projectId}/{frequencySeconds}
- /api/HistoricalArchive/output-points
- /api/Project/output-points-archive-statuses
- /api/Sensors
- /api/Sensors/point-selector
- /api/Sensors/alert-state
- /api/UserDefinedProperty (GET)
- /api/UserDefinedProperty (PUT)
- /api/UserDefinedProperty/remap-userdefinedproperty
- /api/UserDefinedPropertynew-userdefinedproperty
- /api/UserDefinedProperty/id

Headers

After authenticating, calls require a valid authorization token, which is passed in the authentication header. Authentication is provided by the *Identity* and *Token* endpoints. For details about how the user is initially authenticated and how the authorization token is obtained through the API, see [Security](#).

Note: *Identity* and *Token* are not data calls.

Standard HTTP return codes

The API uses the following codes:

- **200 OK**

The **GET** or **POST** transaction processed correctly.

- **206 PARTIAL CONTENT**

Results were too large to return in a single query.

- **401 UNAUTHORIZED**

The account is not yet authenticated or is not in the authenticated users list.

- **403 FORBIDDEN**

The account has authenticated but cannot access the requested resource. This often indicates that a valid user has requested access to a resource (typically a project) but the user's security group does not grant them access to that resource.

- **406 NOT ACCEPTABLE**

The access token has expired and should be refreshed.

For details about obtaining and refreshing a token using the API, see the "Security" chapter.

Date formats

All dates return ISO 8601 date formats. Data is returned in UTC ("Z") time.

For example,

2012-04-23T18:25:43.511Z

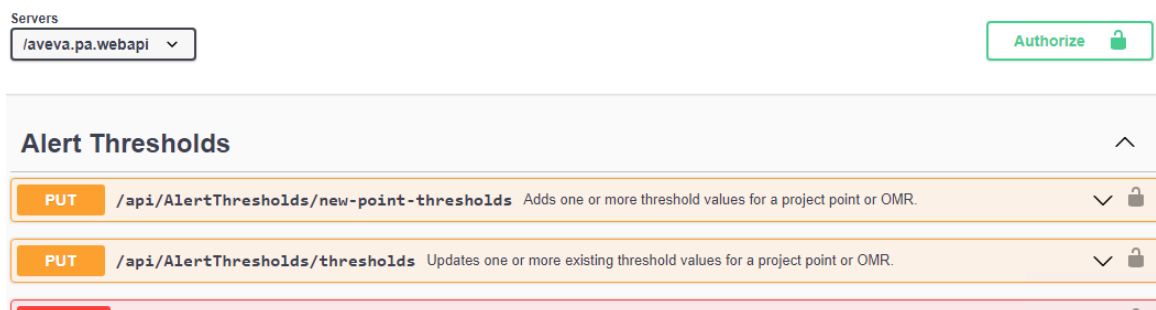
Use the Swagger utility

You can access the Swagger utility through the Web application and then use it to retrieve data and metadata from the AVEVA Predictive Analytics databases.

Open the Swagger utility

- In the Web application, select **API utility and documentation** from the **Help** (?) menu.

The Swagger utility opens in its own browser tab.



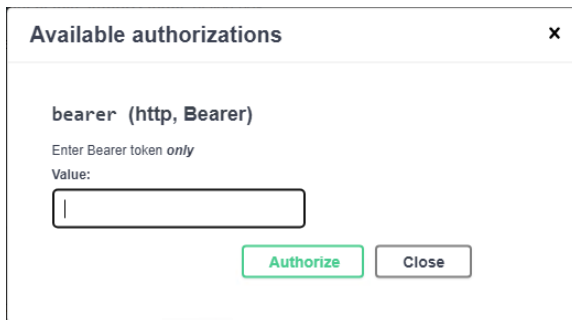
To execute the endpoints, the lock icons must be locked (🔒).

If the lock icons on this screen are unlocked (🔓), then you must authorize with a valid bearer token.

Authorize with a bearer token

1. In the Web application, select **Copy bearer token** from the **Current User** (👤) menu.
2. Return to the Swagger tab.
3. Select any unlocked Lock (🔓) icon, such as the one in the **Authorize** button.

The bearer token is copied to the clipboard and the Available authorizations dialog box opens.



The dialog box titled "Available authorizations" contains the following elements:

- bearer (http, Bearer)**
- Enter Bearer token only*
- Value:**
- A text input field with a vertical cursor.
- Authorize** button (green border)
- Close** button (grey border)

4. Paste the copied bearer token into the **Value** field.
5. Select **Authorize**.

All the lock icons on the page are now locked (🔒) and all endpoints are executable.

Note: Use caution when using APIs for writing data to the database. Improper use of the API may change or delete data unexpectedly. Develop and verify your API applications in a test environment before targeting production data. AVEVA cannot guarantee backward and forward compatibility of APIs from one release to another. You are responsible for verifying your application's compatibility when planning a software upgrade.

Note: Large datasets may take more time to retrieve or modify. Please be patient when working with large datasets.

Chapter 3

Security

API security

AVEVA Predictive Analytics Web REST API implements token-based security. AVEVA Predictive Analytics uses a pair of authorization tokens:

- The short-lived *access* token
This is included in each request and helps determine the authorized user making the request.
- The longer-lived *refresh* token

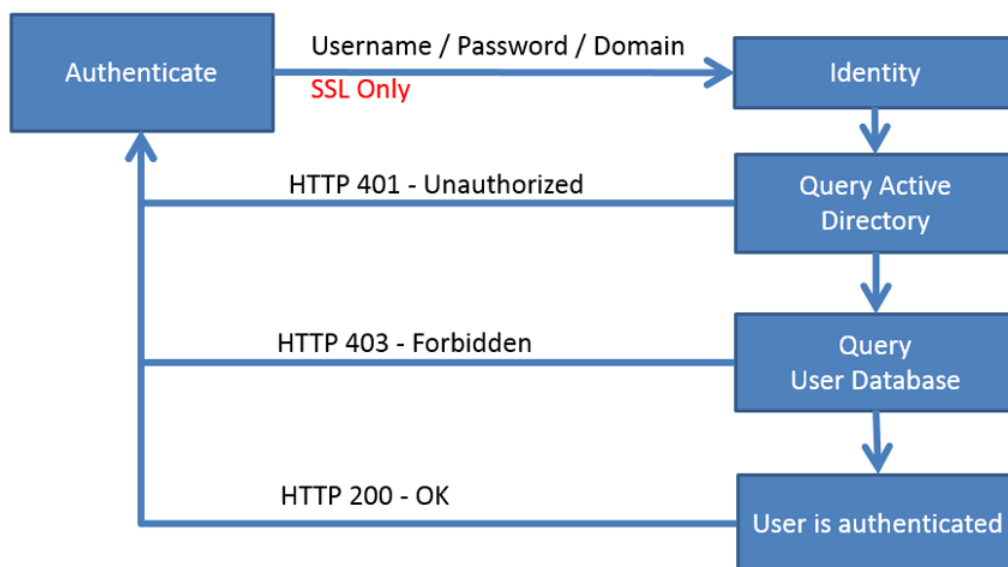
After the access token expires, AVEVA Predictive Analytics can use this to request a new access token without requiring re-authentication.

Security process

AVEVA Predictive Analytics API implements security in a two-part process

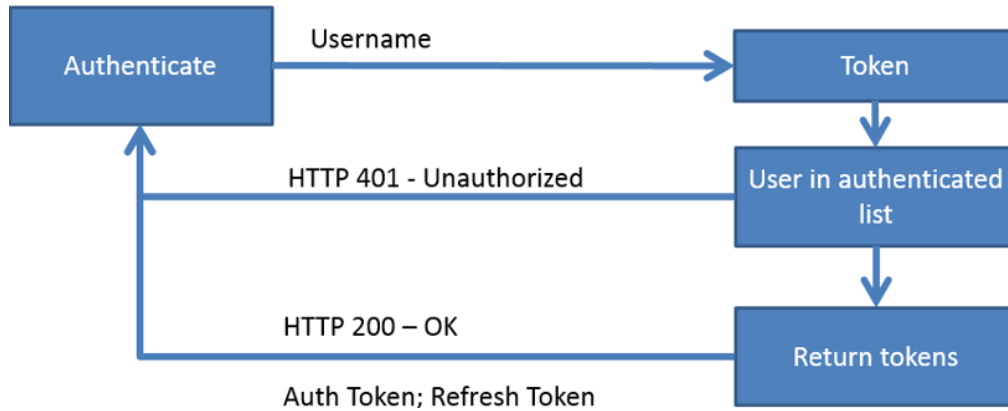
- Authentication (Identity API)

The authentication API uses the username, password and domain and only works for authorized users. This query is only available over SSL.



- Authorization (Token API)

This request returns the authorization and refresh tokens required by the user, as in the token generation process that the browser and AVEVA Predictive Analytics Web use in their session authorizations.



Security calls

There are two security calls:

- Identity is called once to authenticate the user.
- Token is then called to get an access token (or if the current token expires.)

All other calls require the token from the Token call to be sent in the authentication header.

Identity

TITLE	PROJECT
<i>URL</i>	/api/identity
<i>Method</i>	POST
<i>Description</i>	Make this call first when accessing the system programmatically. Identity verifies that users are authorized to use AVEVA Predictive Analytics web. Follow this call with the Token POST request to return tokens to be used for all subsequent data calls.
<i>URL Parameters</i>	None
<i>Header Parameters (64-bit encoded, SSL encrypted)</i>	<i>userid</i> =The user's network domain and account name in the format, DOMAIN\USER <i>password</i> =The password associated with the user's network account
<i>Data Parameters</i>	None

<i>Success Response</i>	Returns if the user is a valid domain user and a valid user. Example: Code: 200 Content: a GUID to be used as the User ID in the token request
<i>Error Response</i>	If the username cannot be authenticated against the Active Directory system, the following returns: Code: 401 UNAUTHORIZED Content: { error : "The account could not be authenticated." } If the user was not granted access in the application or does not have AVEVA Predictive Analytics Analytics Web privileges, the following returns: Code: 403 FORBIDDEN Content: { error : "The account does not have access rights to PRISM." } The call is only available over SSL. A non-SSL request returns a 405 (METHOD NOT ALLOWED) error. Code: 405 METHOD NOT ALLOWED Content: {error: "This resource may only be accessed over SSL."}

Token

TITLE	PROJECT
<i>URL</i>	/token
<i>Method</i>	POST
<i>Description</i>	Make token calls after the identity call and before any data calls. Tokens returned from this call identify the user in subsequent data calls. Also make the token call if the access token expires with an error response of 406 NOT ACCEPTABLE.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	None

<i>Data Parameters</i>	<code>grant_type</code> = "password" This is a fixed value. <code>username</code> = "<guid>" This is the guid returned from a successful Identity call. It can be used once. Generate another guid if another token is required. <code>client_id</code> = "prism-api". This is a fixed value. <code>client_secret</code> = "prism-api". This is a fixed value.
<i>Success Response</i>	Returns if the user previously made a successful request to Identity. Example: Code: 200 Content: <pre>{ "access_token": "H1ts7mDtn81PZxuhKFuU4ZErOxfLty..", "token_type": "bearer", "expires_in": 35999, "refresh_token": "ce9f4d06bdbfb970d127be975974" ... other information }</pre>
<i>Error Response</i>	Returns if the user does not have a valid guid: Code: 400 Bad request Content: <pre>{ "error": "invalid login", "error_description": "Invalid User GUID, could not map to User ID" }</pre>
<i>Sample Call</i>	<pre>{ "grant_type" : "password", "username": "43bb96c1-a8f4-436b-977b-529abceb8c2a", "client_id": "prism-api", "client_secret": "prism-api" }</pre>

Chapter 4

Alert status

Alert status queries

This collection of endpoints provides access to the AVEVA Predictive Analytics asset and project hierarchy, as well as the point configuration for each project. Data from these endpoints includes information about runtime and alert states.

There are three alert endpoints:

- Asset Alert Status
Returns the name and alert state for the requested assets, as well as all child assets and projects assigned to this asset.
- Project Alert Status
Returns basic alert state information about the project identified by the project ID.
- Point Alert Configuration
Returns basic threshold information about the point identified by the PointID, including related output points and thresholds.

Asset alert status

TITLE	PROJECT
URL	/api/assets/ or api/assets/[assetID]
Method	GET
Description	Returns the name and alert state for the requested assets, as well as all child assets and projects assigned to this asset.
URL Parameters	None (Returns the root asset.) AssetID (Returns all assets under the specified asset.)
Header Parameters	Authentication token
Data Parameters	assetID [Integer, optional]

Success Response	Returns if the asset ID is valid. Example: Code: 200 Content: <pre> { "AssetId": 101, "Description": "TEST - HADOOP", "ParentAssetId": 100, "RuntimeState": "Alert", "ChildAssets": [], "ChildProjects": [{ "ProjectId": 3, "TemplateId": 0, "AssetId": 0, "Deployed": true, "AlertWorkFlowStateId": 1, "Name": "TEST - HADOOP", "RunTimeState": "Clear", "RuntimeStateId": 0 }, { "ProjectId": 4, "TemplateId": 0, "AssetId": 0, "Deployed": true, "AlertWorkFlowStateId": 2, "Name": "TEST - HADOOP - WITH ALERTS", "RunTimeState": "Alert", "RuntimeStateId": 2 }, { "ProjectId": 5, "TemplateId": 0, "AssetId": 0, "Deployed": true, "AlertWorkFlowStateId": 2, "Name": "TEST - HADOOP - TAG CHANGE", "RunTimeState": "Clear", "RuntimeStateId": 0 }] }</pre>
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<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." } If the data point does not exist or the user does not have access to it: Code: 403 FORBIDDEN Content: { error : "Point not found or Access Denied" }
<i>Sample Call</i>	GET http://avevapredictiveanalytics/api/assets/100
<i>Notes</i>	Users restricted from accessing a child asset do not see the element in the child asset list. The status of any children do not appear in the rollout.

Project alert status

TITLE	PROJECT
<i>URL</i>	/api/projects/{ProjectID}
<i>Method</i>	GET
<i>Description</i>	Returns basic alert state information about the project identified by the project ID.
<i>URL Parameters</i>	ProjectID [Integer, required] excludeNonModeledPoints [Boolean, optional]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None

<i>Success Response</i>	Example: Code: 200 Content: <pre> { "Points": [{ "ProjectPointId": 1, "PointTypeId": 1, "PointType": "InputSignal", "ProjectId": 1, "Name": "PRISMDEV.CALC.SIN_000", "RunTimeState": "Clear", "AlertWorkFlowStateId": null }, { "ProjectPointId": 2, "PointTypeId": 1, "PointType": "InputSignal", "ProjectId": 1, "Name": "PRISMDEV.CALC.SIN_120", "RunTimeState": "Clear", "AlertWorkFlowStateId": null }, { "ProjectPointId": 3, "PointTypeId": 1, "PointType": "InputSignal", "ProjectId": 1, "Name": "PRISMDEV.CALC.SIN_240", "RunTimeState": "Clear", "AlertWorkFlowStateId": null }], "ProjectId": 1, "TemplateId": 0, "AssetId": 100, "Deployed": true, "AlertWorkFlowStateId": 1, "Name": "TEST - BASIC SINE WAVE", "RunTimeState": "Clear", "RuntimeStateId": 0 }</pre>
<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." } If the data point does not exist or the user does not have access to it: Code: 403 FORBIDDEN Content: { error : "Point not found or Access Denied" }

<i>Sample Call</i>	GET http://avevapredictiveanalytics/api/projects/200
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Point alert configuration

TITLE	PROJECT
<i>URL</i>	/api/points/{PointID}
<i>Method</i>	GET
<i>Description</i>	Returns basic threshold information about the point identified by the PointID, including related output points and thresholds.
<i>URL Parameters</i>	PointID [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None

<i>Success Response</i>	Example: Code: 200 Content: <pre> { "OutputPoints": [{ "ProjectPointId": 22, "Name": "PA.ARCHIVE.00020MR", "PointType": "OutputClusterDistance", "Thresholds": [{ "Type": "HighWarning", "Value": 2 }, { "Type": "HighAlert", "Value": 5 }] }], "Description": "TEST - DEPLOY WHILE SERVER ACTIVE Overall Model Residual", "ExtendedDescription": "TEST - DEPLOY WHILE SERVER ACTIVE TEST - DEPLOY WHILE SERVER ACTIVE Overall Model Residual", "ExtendedId": "TEST - DEPLOY WHILE SERVER ACTIVE TEST - DEPLOY WHILE SERVER ACTIVE Overall Model Residual", "Units": "", "Metric": "", "ProjectPointId": 22, "PointTypeId": 2, "PointType": "OutputClusterDistance", "ProjectId": 2, "Name": "PA.ARCHIVE.00020MR", "RunTimeState": "Alert", "AlertWorkFlowStateId": 24 } </pre>
<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." } If the data point does not exist or the user does not have access to it: Code: 403 FORBIDDEN Content: { error : "Point not found or Access Denied" }
<i>Sample Call</i>	GET http://avevapredictiveanalytics/api/points/301

Notes	Only defined alert thresholds can be returned. The four possible alert attributes are: • High Alert • Low Alert • High Warning • Low Warning
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Chapter 5

History

History queries

Use the HistData history endpoint to retrieve historical time-series results from the Archive Database for a specified project point and time range, including:

- Actual and predicted values
- Overall model residual

HistData

TITLE	PROJECT
URL	/api/histdata/
Method	POST
Description	Returns historical data for the requested point ID.
URL Parameters	None
Header Parameters	Authentication token
Data Parameters	PointIDs [Integer list, required] StartDateTime [UTC Date Time, required] EndDateTime [UTC Date Time, required] FrequencySeconds [Integer, required] The sample interval in seconds

<i>Success Response</i>	Example: Code: 200 Content: <pre>[{ "PointId": 22, "PointName": "PA.ARCHIVE.0002OMR", "StartDateTime": "2019-05-16T09:00:00Z", "EndDateTime": "2019-05-16T11:00:00Z", "FrequencySeconds": 60, "DataPoints": [{ "Value": 1.1434594326490197, "TimeStamp": "2019-05-16T09:00:00Z", "Status": "Good" }, { "Value": 0, "TimeStamp": "2019-05-16T09:01:00Z", "Status": "Good" }, { "Value": 0, "TimeStamp": "2019-05-16T09:02:00Z", "Status": "Good" }, { "Value": 0, "TimeStamp": "2019-05-16T09:03:00Z", "Status": "Good" }] }]</pre>
<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." } If the user has invalid parameters (for example the start date is later than the end date), an error returns describing the invalid parameter. Code: 400 BAD REQUEST Content: { varies } If the point does not exist or the user does not have access to it: Code: 403 FORBIDDEN Content: { error : "Point not found or Access Denied" }

Sample Call	POST http://avevapredictiveanalytics/api/histdata/ Body: <pre>{ "StartDateTime": "2017-07-11T13:00:00Z", "EndDateTime": "2017-07-11T14:30:00Z", "FrequencySeconds": "60", "PointIds": [1,2,3,4,5] }</pre>
Notes	The Point ID must be of a valid input point or OMR as seen during an api/projects/[id] call. The possible status values for each reading are: • <i>Good</i> • <i>Error</i> • <i>Questionable</i> This is used for calculated results where some inputs have a <i>Good</i> status and some have a <i>Bad</i> or <i>Questionable</i> status. This call is all or nothing as it tries to optimize the number of the historical data reads. If any Point ID returns an error, or if the user does not have access to the point, an error returns.

Chapter 6

Alert management

Permitted alert states

The alert management endpoints:

- Provide information about the allowed alert states and supported user-defined values for alert management
- Allow the alert state for project, points, and assets to be updated or cleared

The dashboard initializes with a one-time "look-up" call that provides the allowed states and user-defined values that are used when alerts are cleared. This call only requires one endpoint.

When using the alert management functions, you must tell the calling system:

- Which states are available.
- Which user-defined values are allowed when clearing alerts.

Alert information is exposed through a pair of endpoints that can be called when AVEVA Predictive Analytics Web launches:

- /api/AlarmWorkFlowStates/
- /api/AlarmWorkFlowStates/ClearParameters

The values from these endpoints provide the options that a user (or an AVEVA Predictive Analytics system) can include when calling the POST methods for AlarmWorkFlowState.

Note: Users cannot modify the values in the AlarmWorkFlowStates. There is a small possibility that these values may change between AVEVA Predictive Analytics releases.

The information for the current values is shown in the "Set AlertWorkFlow State" section.

Note: Only system administrators can modify the values of ClearParameters.

These values are only required when user-defined states are passed to the POST method for AlarmWorkFlowState when default alert clear states are not being used.

Alert states

TITLE	PROJECT
URL	/api/AlarmWorkFlowStates/
Method	GET

<i>Description</i>	Returns the list of all alert workflow states defined in the system, along with their text descriptions. These are the states visible in AVEVA Predictive Analytics Web for alert states used to describe the current alert state, such as: • <i>OK/Clear</i> • <i>New Alert</i> • <i>Pending</i> • <i>Sensor Issue</i> The StateID is the key that is required to change an alert state. The other values are informational.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Example: Code: 200 Content: <pre>[{ "Category": 9, "CategoryName": "Clear", "WorkFlowMode": 5, "WorkFlowModeName": "Resolve", "Importance": 0 "ImportanceName": "Normal" "Ranking": 1 "StateID": 1 "UserCanSelect": true "AssetRanking": 29 }, { "Category": 1, "CategoryName": "ActionDue", "WorkFlowMode": 4, "WorkFlowModeName": "Act" "Importance": 1 "ImportanceName": "Increased" "Ranking": 2 "StateID": 2 "UserCanSelect": false "AssetRanking": 1 }, (the rest of the data)]</pre>

<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." }
<i>Sample Call</i>	GET http://avevapredictiveanalytics/api/AlarmWorkFlowStates/
<i>Notes</i>	The following numeric values have paired values that include a text string: • Category • Workflowmode • Importance The text matches what is shown in AVEVA Predictive Analytics Web. The stateID is the unique identifier used to set state value. This number is passed to the API to change the alert state of an asset, project, or point. Ranking and AssetRanking determine the priority for rolling up the alert states. Ranking determines the state of the project. AssetRanking determines the state of the asset. The UserCanSelect property indicates if this is a valid state that the API can set.

Clear parameters

TITLE	PROJECT
<i>URL</i>	/api/AlarmWorkFlowStates/ClearParameters
<i>Method</i>	GET
<i>Description</i>	Returns a list of the allowed user-defined categorizations that can be passed when an alert is cleared
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None

<i>Success Response</i>	Example: Code: 200 Content: <pre>{ "ClearTypes": [{ "Id": 1, "Value": "Mechanical Problem" }, { "Id": 2, "Value": "Electrical Problem" } (the rest of the data)], "ActionTypes": [{ "Id": 11, "Value": "Contact Plant" }, { "Id": 12, "Value": "Contact SME" } (the rest of the data)], "ClassificationTypes": [{ "Id": 23, "Value": "Environmental Issue" }, { "Id": 24, "Value": "Operational Issue" } (the rest of the data)] }</pre>
<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." }
<i>Sample Call</i>	GET http://avevapredictiveanalytics/api/AlarmWorkFlowStates/ClearParameters
<i>Notes</i>	There is a list of key value pairs returned for each list visible in AVEVA Predictive Analytics Web when a user clears an alert. When posting a <i>clear</i> type alert workflow state change (StateID = 1), the corresponding IDs must be provided for each alert classification states.

Set AlarmWorkflow state

Three API endpoints can change the state of an alert. All three take the same parameter object. But they operate on different endpoints. The calls are identical, but the primary key differs:

- asset ID
- project ID
- point ID

TITLE	PROJECT
<i>URL</i>	/api/assets/{id}/AlarmWorkFlowState /api/projects/{id}/AlarmWorkFlowState /api/points/{id}/AlarmWorkFlowState
<i>Method</i>	POST
<i>Description</i>	Sets the alert workflow state for the specified resource. This may be an: • Asset • Project • Point
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	<i>State ID</i> [Integer, required] <i>ExpirationDateUtc</i> [UTC Date Time, required and used for <i>pending</i> states] <i>Notes</i> [string, optional and used for <i>clear</i> states] <i>ClearTypeID</i> [Integer, required and used only for <i>clear</i> states] <i>ActionTypeID</i> [Integer, required and used only for <i>clear</i> states] <i>ClassificationTypeID</i> [Integer, required and used only for <i>clear</i> states]
<i>Success Response</i>	Example: Code: 200 Content: {none}

<i>Error Response</i>	If the user does not have a valid token: Code: 406 Not Acceptable Content: { error : "Unauthenticated Token." } If the user has invalid parameters (such as not supplying a parameter required for the specified state), an error returns, describing the invalid parameter. Code: 400 BAD REQUEST Content: { varies} If the user does not have permissions to change the alarm state: Code: 403 FORBIDDEN Content: { error : "User is not allowed to clear alarms for this project."}
<i>Sample Call</i>	POST http://avevapredictiveanalytics/api/api/points/200/AlarmWorkFlowState Body: { "StateID": 1, "Notes": "Alarm cleared from dashboard", "ClearTypeID": 1, "ActionTypeID": 12, "ClassificationTypeID": 24 "ExpirationDateUtc": "05/18/2019 12:00:00:00" }
<i>Notes</i>	The clear state requires the following parameters: • <i>cleartypeid</i> • <i>actiontypeid</i> • <i>classificationtypeid</i> A note is optional. These parameters are only used in this scenario. Pending and deferred action transactions require the <i>ExpirationDateUtc</i> parameter, which is only used in this scenario.

Alert state codes and icons

The alert state is determined by a State ID. The following table displays the State IDs and their corresponding icons and definitions. Icon graphic and color are determined by the category, mode, and importance of the alert, which are managed by AVEVA Predictive Analytics Web users selecting from drop-down menus.

Icon	State ID	Category Name	Mode Name	Importance Name	Category ID	Mode ID	Importance
	9	ACKNOWLEDGE	REVIEW	INCREASED	4	3	1
	10	ACKNOWLEDGE	REVIEW	NORMAL	4	3	0
	11	ACKNOWLEDGE	WAIT	NORMAL	4	2	0
	2	ACTIONDUE	ACT	INCREASED	1	4	1
	1	CLEAR	RESOLVE	NORMAL	9	5	0
	5	EQUIPMENT	ACT	INCREASED	3	4	1
	6	EQUIPMENT	ACT	NORMAL	3	4	0
	21	EQUIPMENT	IGNORE	NORMAL	3	1	0
	3	EQUIPMENT	REVIEW	INCREASED	3	3	1
	4	EQUIPMENT	REVIEW	NORMAL	3	3	0
	18	EQUIPMENT	WAIT	NORMAL	3	2	0
	14	MODEL	ACT	INCREASED	7	4	1
	15	MODEL	ACT	NORMAL	7	4	0
	22	MODEL	IGNORE	NORMAL	7	1	0
	12	MODEL	REVIEW	INCREASED	7	3	1
	13	MODEL	REVIEW	NORMAL	7	3	0
	19	MODEL	WAIT	NORMAL	7	2	0
	7	MONITOR	REVIEW	INCREASED	5	3	1
	8	MONITOR	REVIEW	NORMAL	5	3	0
	24	NEW	REVIEW	INCREASED	2	3	1
	25	NEW	REVIEW	NORMAL	2	3	0
	23	PENDING	IGNORE	NORMAL	6	1	0
	16	PENDING	REVIEW	INCREASED	6	3	1
	17	PENDING	REVIEW	NORMAL	6	3	0
	20	PENDING	WAIT	NORMAL	6	2	0
	26	SENSOR	ACT	INCREASED	8	4	1
	27	SENSOR	ACT	NORMAL	8	4	0
	29	SENSOR	IGNORE	NORMAL	8	1	0
	28	SENSOR	WAIT	NORMAL	8	2	0
	0	UNKNOWN	UNKNOWN	NORMAL	0	0	0

Chapter 7

Alert thresholds

Add Threshold

TITLE	PROJECT
URL	/api/AlertThresholds/new-point-thresholds
Method	PUT
Description	Adds one or more threshold values for a project point or OMR.
URL Parameters	None
Header Parameters	Authentication token
Data Parameters	Project Point ID [Integer, required] Input Project Point Type [Integer, optional] Alert Threshold Type [Integer, required] Threshold Value [Number, optional] Persistence Denomination [Integer, required] Note [String, optional] Minimum Count [Integer, required] Seconds or Event Count [Integer, required] Is Critical [Boolean, optional]
Success Response	Code: 200. Record(s) created.

<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • The request must contain a unique combination of project point and alert threshold type. Duplicate records are prohibited. • Could not find project point or project. • Threshold type is not valid for project point type. • Project is not deployed. • The ratio of Minimum to Out of total persistence fields must be greater than 0 and no greater than one. • Could not add a record to project history. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator. Code: 423, Locked - Project is locked.
<i>Notes</i>	A threshold is a cutoff value for a filter or an alert. Use this call to create one or more thresholds for a point or OMR. This also adds details to project history.

Update Threshold

TITLE	PROJECT
<i>URL</i>	/api/AlertThresholds/thresholds
<i>Method</i>	PUT
<i>Description</i>	Updates one or more existing threshold values for a project point or OMR.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token

<i>Data Parameters</i>	Threshold ID [Integer, required] Is Active [Boolean, optional] Alert Threshold Type [Integer, required] Threshold Value [Number, optional] Persistence Denomination [Integer, required] Note [String, optional] Minimum Count [Integer, required] Seconds or Event Count [Integer, required] Is Critical [Boolean, optional]
<i>Success Response</i>	Code: 200. Record(s) updated.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • Invalid threshold ID. • The request must contain a unique combination of project point and alert threshold type. Duplicate records are prohibited. • Could not find project point or project. • Threshold type is not valid for project point type. • Project is not deployed. • The ratio of Minimum to Out of total persistence fields must be greater than 0 and no greater than one. • Could not add a record to project history. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator. Code: 423, Locked - Project is locked.
<i>Notes</i>	A threshold is a cutoff value for a filter or an alert. Use this call to change an existing threshold for a point or OMR.

Delete Threshold

TITLE	PROJECT
<i>URL</i>	/api/AlertThresholds/thresholds
<i>Method</i>	DELETE

<i>Description</i>	Deletes one or more threshold values.
<i>URL Parameters</i>	thresholdIds [Array (integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record(s) deleted.
<i>Notes</i>	A threshold is a cutoff value for a filter or an alert. Use this call to permanently remove one or more thresholds from a point or OMR. A successful delete cannot be undone.

Restore Threshold

TITLE	PROJECT
<i>URL</i>	/api/AlertThresholds/template-references-reset
<i>Method</i>	POST
<i>Description</i>	Restores template-based project thresholds to the template threshold values.
<i>URL Parameters</i>	thresholdIds [Array (integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record(s) reset.
<i>Error Response</i>	Code: 204, No Action - The project is not template-based. Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid threshold. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator. Code: 423, Locked - Project is locked.

<i>Notes</i>	A threshold is a cutoff value for a filter or an alert. Projects, by default, inherit threshold values from their parent templates, but the project values can be changed. Use this call to modify the project's threshold values to the original template reference values. This also adds details to project history.
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Retrieve Threshold by Threshold ID

TITLE	PROJECT
<i>URL</i>	/api/AlertThresholds/threshold/{thresholdId}
<i>Method</i>	GET
<i>Description</i>	Retrieves threshold information based on threshold ID.
<i>URL Parameters</i>	thresholdId [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid threshold. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	A threshold is a cutoff value for a filter or an alert. Use this call to return threshold values for a point, OMR, or template.

Retrieve Threshold by Point ID

TITLE	PROJECT
<i>URL</i>	/api/AlertThresholds/threshold/{pointId}
<i>Method</i>	GET
<i>Description</i>	Retrieves threshold information based on the point ID.
<i>URL Parameters</i>	pointId [Integer, required]

<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record(s) returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Point not found. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	A threshold is a cutoff value for a filter or an alert. Use this call to retrieve threshold values for a point, OMR, or template.

Chapter 8

Audit

Get Users

TITLE	PROJECT
URL	/api/Audit/Get-Audit-Lookup-Users
Method	GET
Description	Returns the list of users who have records in audit history.
URL Parameters	None
Header Parameters	Authentication token
Data Parameters	None
Success Response	Example: Code: 200 Content: <pre>[{ "displayText": "string", "value": "string" }]</pre>
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Sample Call	GET http://avevapredictiveanalytics/api/Audit/Get-Audit-Lookup-Users
Notes	

Get Auditing Categories

TITLE	PROJECT
URL	/api/Audit/Get-Audit-Lookup-EntityTypes
Method	GET
Description	Returns the list auditing categories.
URL Parameters	None
Header Parameters	Authentication token
Data Parameters	None
Success Response	Example: Code: 200 Content: <pre>[{ "displayText": "string", "value": "string" }]</pre>
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Sample Call	GET http://avevapredictiveanalytics/api/Audit/Get-Audit-Lookup-EntityTypes
Notes	Entity types include alerts, projects, templates, etc. that are audited in AVEVA Predictive Analytics.

Get Audit History

TITLE	PROJECT
URL	/api/Audit/Get-Audit-Grid-Data
Method	GET
Description	Returns the audit records.

<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	AuditEntityTypeEnum [Integer] ("-1" for all entity types) UserSid [String] ("-1" for all users) StartDate [Date-Time String] (Empty string or null for no start date) EndDate [Date-Time String] (Empty string or null for no end date) SearchText [String] (Empty string or null for no search text) PageNumber [Integer] (First page of data is 1) RowsPerPage [Integer]
<i>Success Response</i>	Example: Code: 200 Content: <pre>{ "rows": [{ "auditHeaderId": 0, "headerChangedByUser": "string", "headerChangeOnDate": "2022-11-03T20:18:50.177Z", "headerSummaryText": "string", "detailSummaryText": "string", "detailAuditText": "string" }], "countTotalRecords": 0 }</pre>
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Sample Call</i>	GET http://avevapredictiveanalytics/api/Audit/Get-Audit-Grid-Data
<i>Notes</i>	

Chapter 9

Calculation points

Retrieve Calculation Point

TITLE	PROJECT
<i>URL</i>	/api/CalculationPoints
<i>Method</i>	GET
<i>Description</i>	Retrieves information about existing calculation points.
<i>URL Parameters</i>	ids [Array (integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record(s) returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	A calculation point is a project point, designed in the equation editor, whose value is derived from a mathematical formula built from other values in the project. Use this call to retrieve information about one or more calculation points based on their point IDs.

Update Calculation Point

TITLE	PROJECT
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<i>URL</i>	/api/CalculationPoints
<i>Method</i>	PUT
<i>Description</i>	Updates one or more existing calculation points.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	Description [String, optional] Digital Group ID [Integer, optional] Extended Description [String, optional] Extended ID [String, optional] ID [Integer, required] Units [String, optional]
<i>Success Response</i>	Code: 200. Record(s) updated.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • Could not get real-time service information. • Invalid calc point. • Calculation point is template-derived and cannot be updated. • Calculation point is an elapsed time and cannot be edited. • Invalid digital point ID. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator. Code: 423, Locked - Project is locked.
<i>Notes</i>	A calculation point is a project point, designed in the equation editor, whose value is derived from a mathematical formula built from other values in the project. Use this call to change information about one or more calculation points.

Chapter 10

Digital point groups

Retrieve Digital Point Groups

TITLE	PROJECT
URL	/api/DigitalGroups
Method	GET
Description	Retrieves a list of digital point groups.
URL Parameters	None
Header Parameters	Authentication token
Data Parameters	None
Success Response	Code: 200. Record(s) returned.
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Sample Call	https://avevapredictiveanalytics/api/DigitalGroups
Notes	A digital point group is a collection of digital point labels (representations of project point state values) that AVEVA Predictive Analytics can use for a project point. Use this call to retrieve the list of digital point groups.

Chapter 11

Fault diagnostics

Retrieve Fault Diagnostic

TITLE	PROJECT
URL	/api/FaultDiagnostic
Method	GET
Description	Retrieves description, next step, and user defined properties for a fault.
URL Parameters	faultDiagnosticId [Integer, required]
Header Parameters	Authentication token
Data Parameters	None
Success Response	Code: 200. Record returned.
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Notes	Fault diagnostics help capture engineering knowledge about familiar mechanical failures for different types of equipment. Fault diagnostics are defined in templates to make them available to all projects that reference that template. Use this call to retrieve the description, next steps advice, and user defined properties information for the fault, based on the fault ID.

Retrieve Fault Diagnostic Based on ID, Frequency, Time Span

TITLE	PROJECT
<i>URL</i>	/api/FaultDiagnostic/project/{projectId}/{frequencySeconds}
<i>Method</i>	GET
<i>Description</i>	Retrieves information about a fault based on: • ID • sample frequency • time span
<i>URL Parameters</i>	projectId [Integer, required] frequencySeconds [Integer, required] startDate [String (date-time), required] endDate [String (date-time), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record(s) returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • More than five project IDs entered. • Start date must precede end date. • Project not found. • Project is not template-based. • Project ID is a template. • Project does not archive OMR, which is required to process faults. • Could not retrieve project time-series data. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Fault diagnostics helps capture engineering knowledge about familiar mechanical failures for different types of equipment. Faults are defined in templates to make them available to all projects that reference that template. Use this call to retrieve the average and recent fault matches.

Retrieve Fault Diagnostic Based on ID, Frequency, Recent Match, Time Span

TITLE	PROJECT
<i>URL</i>	/api/FaultDiagnostic/project/{projectId}/{frequencySeconds}/{recentMatchPercentage}
<i>Method</i>	GET
<i>Description</i>	Retrieves information about a fault based on: • ID • sample frequency • recent match • time span
<i>URL Parameters</i>	projectId [Integer, required] frequencySeconds [Integer, required] recentMatchPercentage [Number, required] startDate [String (date-time), required] endDate [String (date-time), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record(s) returned.

<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • More than five project IDs entered. • Start date must precede end date. • Project not found. • Project is not template-based. • Project ID is a template. • Project does not archive OMR, which is required to process faults. • Could not retrieve project time-series data. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Fault diagnostics helps capture engineering knowledge about familiar mechanical failures for different types of equipment. Faults are defined in templates to make them available to all projects that reference that template. Use this call to retrieve the average and recent fault matches.

Retrieve Fault Diagnostic Based on Project

Fault diagnostics help capture engineering knowledge about familiar mechanical failures for different types of equipment. Faults are defined in templates to make them available to all projects that reference that template. Use this call to retrieve information about up to five projects' faults.

TITLE	PROJECT
<i>URL</i>	/api/FaultDiagnostic/fault-details
<i>Method</i>	POST
<i>Description</i>	Retrieves information about fault diagnostics for a project.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	Project IDs [Integer, optional] Frequency Seconds [Integer, optional] Start Date [String (date-time), required] End Date [String (date-time), required]
<i>Success Response</i>	Code: 200. Record updated.

<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • More than five project IDs entered. • Start date must be precede end date. • Project not found. • Project is not template-based. • Project ID is a template. • Project does not archive OMR, which is required to process faults. • Could not retrieve project time-series data. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Fault diagnostics helps capture engineering knowledge about familiar mechanical failures for different types of equipment. Faults are defined in templates to make them available to all projects that reference that template. Fault signatures are rules that observe deviations in project point values. Use this call to retrieve faults and fault signature information.

Retrieve Fault Diagnostic and Fault Signatures Based on ID, Frequency, Recent Match, Time Span

TITLE	PROJECT
<i>URL</i>	/api/FaultDiagnostic/summary
<i>Method</i>	GET
<i>Description</i>	Retrieves information about a fault and its fault signatures based on: • ID • sample frequency • recent match • time span

<i>URL Parameters</i>	ProjectId [Integer, required] FaultIds [Array (integer), required] FrequencySeconds [Integer, required] RecentMatchPercentage [Number, required] StartDate [String (date-time), required] EndDate [String (date-time), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • More than five project IDs entered. • Start date must be precede end date. • Project not found. • Project is not template-based. • Project ID is a template. • Project does not archive OMR, which is required to process faults. • Could not retrieve project time-series data. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Fault diagnostics helps capture engineering knowledge about familiar mechanical failures for different types of equipment. Faults are defined in templates to make them available to all projects that reference that template. Fault signatures are rules that observe deviations in project point values. Use this call to retrieve faults and fault signature information.

Chapter 12

Forecast

Retrieve Forecast Values

TITLE	PROJECT
URL	/api/Forecast
Method	POST
Description	Retrieves information about future value prediction.
URL Parameters	None
Header Parameters	Authentication token

<i>Data Parameters</i>	Fitted Values • Predictions Start [String (date-time), optional] End [String (date-time), optional] Record Count [Integer, optional] • Upper Limit Start [String (date-time), optional] End [String (date-time), optional] Record Count [Integer, optional] • Lower Limit Start [String (date-time), optional] End [String (date-time), optional] Record Count [Integer, optional] Forecast Values • Predictions Start [String (date-time), optional] End [String (date-time), optional] Record Count [Integer, optional] • Upper Limit Start [String (date-time), optional] End [String (date-time), optional] Record Count [Integer, optional] • Lower Limit Start [String (date-time), optional] End [String (date-time), optional] Record Count [Integer, optional] Project Point ID [Integer, optional]
<i>Success Response</i>	Code: 200. Record returned.

<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Use this call to retrieve predictions about future values of project points based on past behavior and current conditions.

Retrieve Deployment Parameters

TITLE	PROJECT
<i>URL</i>	/api/Forecast/deploymentParameters/{forecastModelId}
<i>Method</i>	GET
<i>Description</i>	Retrieves deployment parameters for a forecast model.
<i>URL Parameters</i>	ForecastModelId [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Use this call to retrieve predictions about future values of project points based on past behavior and current conditions.

Save Deployment Parameters

TITLE	PROJECT
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<i>URL</i>	/api/Forecast/deploymentParameters/{projectId}
<i>Method</i>	POST
<i>Description</i>	Saves project point deployment parameters.
<i>URL Parameters</i>	ProjectPointId [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Use this call to save deployment parameters for a given project Point.

Delete Forecast Model

TITLE	PROJECT
<i>URL</i>	/api/Forecast/forecastModelId
<i>Method</i>	DELETE
<i>Description</i>	Deletes a forecast model.
<i>URL Parameters</i>	ForecastModelId [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.

<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Use this call to delete an existing model. A successful delete cannot be undone.

Retrain Forecast Model

TITLE	PROJECT
<i>URL</i>	/api/Forecast/retrainForecastModel/{forecastModelId}
<i>Method</i>	POST
<i>Description</i>	Retrains a forecast model and updates the project point deployment parameters
<i>URL Parameters</i>	ForecastModelId [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Use this call to retrain a forecast model and update the deployment parameters for a given project point.

Retrieve Time Remaining

TITLE	PROJECT
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<i>URL</i>	/api/Forecast/deployed-mode-results
<i>Method</i>	GET
<i>Description</i>	Retrieves forecast deployed model from the Archive Database.
<i>URL Parameters</i>	ForecastModelId [Integer, required] evaluatedTime [String] totalDataPionts [Integer]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	The results of the deployed forecast model calculations are stored in the Archive Database. Use this call to retrieve deployed forecast model point time-series data for a project point.

Retrieve Forecast Model

TITLE	PROJECT
<i>URL</i>	/api/Forecast/deployed-model-results
<i>Method</i>	GET
<i>Description</i>	Retrieves forecast deployed model from the Archive Database.
<i>URL Parameters</i>	ForecastModelId [Integer, required] evaluatedTime [String] totalDataPoints [Integer]
<i>Header Parameters</i>	Authentication token

<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Invalid fault ID. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	The results of the deployed forecast model calculations are stored in an Archive Database. Use this call to retrieve deployed forecast model point time-series data for a project point.

Chapter 13

Historian points

Retrieve Historian Points

TITLE	PROJECT
<i>URL</i>	/api/HistorianPoints
<i>Method</i>	GET
<i>Description</i>	Retrieves historian points.
<i>URL Parameters</i>	ids [Array (integer), optional]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record{s} returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. At least one historian point ID is invalid. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for point. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	A historian point is a project point actively linked to an online historian. Use this call to retrieve a list of one or more historian points based on their ID(s).

Change Historian Points

TITLE	PROJECT
<i>URL</i>	/api/HistorianPoints

<i>Method</i>	PUT
<i>Description</i>	Changes historian points.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	Flatline Time Minutes [Integer, optional] High Out of Range Limit [Number, optional] Low Out of Range Limit [Number, optional] Manual Override [Boolean, optional] Recovery Time Minutes [Integer, optional] Status [Boolean, optional] Description [String, optional] Digital Group ID [Integer, optional] Extended Description [String, optional] Extended ID [String, optional] ID [Integer, required] Units [String, optional]
<i>Success Response</i>	Code: 200. Record updated.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. - Invalid historian point. - High out of range value must exceed low out of range value. - Invalid digital point ID parameter. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for point. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	A historian point is a project point actively linked to an online historian. Use this call to change one or more existing historian points.

Chapter 14

Historical archive

Retrieve OMR

The results of operational profile calculations are stored in the Archive Database. These output points and OMRs record historical information. Use this call to retrieve OMR time-series data for a project.

TITLE	PROJECT
URL	/api/HistoricalArchive/omr/{projectId}/{frequencySeconds}
Method	GET
Description	Retrieves OMR historical data.
URL Parameters	projectId [Integer, required] frequencySeconds [Integer, required] startDate [String (date-time), required] endDate [String (date-time), required]
Header Parameters	Authentication token
Data Parameters	None
Success Response	Code: 200. Record returned.
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Notes	The results of operational profile calculations are stored in the Archive Database. These output points and OMRs record historical information. Use this call to retrieve output point time-series data for a project.

Retrieve Output Points

TITLE	PROJECT
<i>URL</i>	/api/HistoricalArchive/output-points
<i>Method</i>	POST
<i>Description</i>	Retrieves output point history.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	Frequency Seconds [Integer, required] Start Date [String (date-time), required] End Date [String (date-time), required] Input Point IDs [Integer, required] Output Point Types [Integer, required]
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	The results of operational profile calculations are stored in the Archive Database. These output points and OMRs record historical information. Use this call to retrieve output point time-series data for a project.

Projects

Retrieve Output Points Archive Statuses

TITLE	PROJECT
URL	/api/Project/output-points-archive-statuses
Method	GET
Description	Retrieves output points based on project ID.
URL Parameters	projectIds [Array (integer), required]
Header Parameters	Authentication token
Data Parameters	None
Success Response	Code: 200. Record returned.
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Notes	An output point is the result of calculations by an operational profile and then recorded in the Archive Database. Use this call to retrieve the types of output points for a project..

Retrieve Training Dataset by Project

TITLE	PROJECT
URL	/api/Project

<i>Method</i>	GET
<i>Description</i>	Retrieves training dataset info based on project ID.
<i>URL Parameters</i>	projectId [(integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized - User token is invalid. - User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	Use this call to retrieve information about a training dataset.

Retrieve Training Dataset by Dataset

TITLE	PROJECT
<i>URL</i>	/api/Project/training-dataset
<i>Method</i>	GET
<i>Description</i>	Retrieves training dataset information based on dataset ID.
<i>URL Parameters</i>	dataSetId [(integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.

Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges for project. Code: 403, Forbidden - Contact system administrator.
Notes	Use this call to retrieve information about a training dataset.

Chapter 16

Sensors

Retrieve Sensors

TITLE	PROJECT
URL	/api/Sensors
Method	POST
Description	Retrieves information on sensors.
URL Parameters	None
Header Parameters	Authentication token
Data Parameters	ID [Integer, required] Results Type [Integer, required] Results Format [Array (integer), required] Filters [Integer, optional]
Success Response	Code: 200. Record returned.
Error Response	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized - User token is invalid. - User lacks privileges. Code: 403, Forbidden - Contact system administrator.
Notes	The Sensors screen in AVEVA Predictive Analytics Web displays alerts based on problems with sensor instruments. Use this call to retrieve a list of points with sensor-based alerts in either list or in summary format.

Retrieve Sensors in Alert

TITLE	PROJECT
<i>URL</i>	/api/Sensors/point-selector
<i>Method</i>	GET
<i>Description</i>	Retrieves a list of sensors in alert.
<i>URL Parameters</i>	ids [Array (integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	The Sensors screen in AVEVA Predictive Analytics Web displays alerts based on problems with sensor instruments. Use this call to retrieve a list of points.

Change Sensor Alert State

TITLE	PROJECT
<i>URL</i>	/api/Sensors/alert-state
<i>Method</i>	POST
<i>Description</i>	Changes a sensor alert state.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	IDs [Integer, required] Alert Managed State [Integer, required] Defer Date UTC [String, optional] Assigned to User ID [String, optional] Update Assign To [Boolean, optional]

<i>Success Response</i>	Code: 200. Record returned.
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized • User token is invalid. • User lacks privileges. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	The Sensors screen in AVEVA Predictive Analytics Web displays alerts based on problems with sensor instruments. Use this call to change an alert state.

Chapter 17

User defined property details

Retrieve User Defined Property Details

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedProperty
<i>Method</i>	GET
<i>Description</i>	Retrieves user defined property details.
<i>URL Parameters</i>	useDefinedPropertyTypeIds [array (integer), required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200, Record(s) updated.
<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. At least one user defined property type is required. Code: 401, Unauthorized. • User token is invalid. • User lacks privileges. Code: 403 Forbidden, Contact system administrator.
<i>Notes</i>	User Defined Properties are customized in-application parameters and values for personal or shared use. User defined property details are grouped under user defined property types. Use this call to retrieve the details for one or more types based on the type's ID(s).

Change User Defined Property Detail

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedProperty
<i>Method</i>	PUT

<i>Description</i>	Changes one or more user defined property details.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	ID [Integer, optional] Name [String, optional] Is Enabled [Boolean, required] Display Order [Integer, required] Numeric Value [Number, optional] Text Value [String, optional]
<i>Success Response</i>	Code: 200, Record(s) updated.
<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. - Invalid user defined property detail. - Duplicate user defined property detail. Code: 401, Unauthorized. - User token is invalid. - User lacks privileges. Code: 403 Forbidden, Contact system administrator.
<i>Notes</i>	User defined property details are customized in-application parameters and values for personal or shared use. User defined property details are grouped under user defined property types. Use this call to modify information about a user defined property detail.

Remap User Defined Property Detail

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedProperty/remap-userdefinedproperty
<i>Method</i>	PUT
<i>Description</i>	Remaps user defined property dependencies to another detail.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	User Defined Property ID [Integer, optional] Remap To User Defined Property ID [Integer, optional]
<i>Success Response</i>	Code: 200, Record(s) updated.

<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. • Invalid remap due to type mismatch or invalid detail ID. • Invalid user defined property detail. • Cannot remap detail to a closed case Code: 401, Unauthorized. • User token is invalid. • User lacks privileges. Code: 403 Forbidden, Contact system administrator.
<i>Notes</i>	User defined properties are customized in-application parameters and values for personal or shared use. A user defined property type is configured as a list of details, a text field, a numeric field, or a calculation. If the type configuration is changed (for example) from text to numeric or list to text, then the detail dependencies must be remapped to a replacement detail. Use this call to modify the detail dependency. Remapping may lose data.

Create User Defined Property Detail

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedProperty/new-userdefinedproperty
<i>Method</i>	POST
<i>Description</i>	Creates a user defined property detail
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	User Defined Property Type ID System Classification [Integer, required] ID [Integer, optional] Name [String, required] Is Enabled [Boolean, required] Display Order [Integer, required] Numeric Value [Number, optional] Text Value [String, optional]
<i>Success Response</i>	Code: 200, Record(s) successfully created.

<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. • Invalid user defined property detail. • Duplicate user defined property detail. Code: 401, Unauthorized. • User token is invalid. • User lacks privileges. Code: 403 Forbidden, Contact system administrator.
<i>Notes</i>	User defined properties are customized in-application parameters and values for personal or shared use. A user defined property type is configured as a list of details, a text field, a numeric field, or a calculation. If the type configuration is changed (for example) from text to numeric or list to text, then the detail dependencies must be remapped to a replacement detail. Use this call to create a new detail under an existing type.

Delete User Defined Property Detail

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedProperty/id
<i>Method</i>	DELETE
<i>Description</i>	Deletes a user defined property detail.
<i>URL Parameters</i>	Id [Integer, required]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200, Record deleted.
<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. • Invalid detail ID. • Detail is in use and cannot be deleted. • Detail is the only remaining one in its type Code: 401, Unauthorized. • User token is invalid. • User lacks privileges. Code: 403 Forbidden, Contact system administrator.

Notes	User Defined Properties are customized in-application parameters and values for personal or shared use. User defined property details are grouped under user defined property types. Use this call to delete an existing detail. A successful delete cannot be undone. Some reserved user defined property details cannot be deleted.
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Chapter 18

User defined property types

Retrieve User Defined Property Types

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedPropertyType
<i>Method</i>	GET
<i>Description</i>	Retrieves information about user defined property types.
<i>URL Parameters</i>	categoryId [Integer, optional]
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200
<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. Code: 401, Unauthorized. • User token is invalid. • User lacks privileges. Code: 403 Forbidden, Contact system administrator.
<i>Notes</i>	User defined properties are customized in-application parameters and values for personal or shared use. User defined property types are grouped in categories. Use this call to retrieve all the user defined types for one or all categories.

Change User Defined Property Type

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedPropertyType
<i>Method</i>	POST

<i>Description</i>	Changes an existing user defined property type.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	ID [Integer, required] Name [String, required] Description [String, optional] IsEnabled [Boolean, optional] CalcEquation [String, optional] TypeConfig [Integer, required]
<i>Success Response</i>	Code: 200
<i>Error Response</i>	Code: 400 BadRequest - Validation error or system error. See log file for details. - Invalid type ID - Name and description cannot be modified for a reserved type. - Name is already used by a reserved type. - Cannot delete a type used by another type's calculation. - Text value for type is already a reserved type name. - Calculation is invalid. Code: 401, Unauthorized. - User token is invalid. - User lacks privileges. Code: 403 Forbidden, Contact system administrator.
<i>Notes</i>	User Defined Properties are customized in-application parameters and values for personal or shared use. User defined property types are grouped in categories. Use this call to change a type. The update may delete existing associations to user defined property details.

Delete User Defined Property Type

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedPropertyType
<i>Method</i>	DELETE
<i>Description</i>	Deletes a user defined property type.
<i>URL Parameters</i>	userDefinedPropertyTypeId [Integer, required]

<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	None
<i>Success Response</i>	Code: 200, Record deleted
<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. - Invalid type ID - Type is used by a calculation. Code: 401, Unauthorized - User token is invalid. - User lacks privileges. Code: 403, Forbidden - Contact system administrator
<i>Notes</i>	User Defined Properties are customized in-application parameters and values for personal or shared use. User defined property types are grouped in categories. Use this call to delete a type. A successful delete cannot be undone. If the type has details, then the details must be remapped to another user defined property type.

Create User Defined Property Type

TITLE	PROJECT
<i>URL</i>	/api/UserDefinedPropertyType/new-userdefinedpropertytype
<i>Method</i>	POST
<i>Description</i>	Creates a user defined property type.
<i>URL Parameters</i>	None
<i>Header Parameters</i>	Authentication token
<i>Data Parameters</i>	<i>System Classification</i> [Integer, required] <i>Category ID</i> [Integer, required] <i>Name</i> [String, required] <i>Description</i> [String, optional] <i>IsEnabled</i> [Boolean, optional] <i>Calc Equation</i> [String, optional] <i>Type Configuration</i> [Integer, required]
<i>Success Response</i>	Code: 201. Record successfully created

<i>Error Response</i>	Code: 400, BadRequest - Validation error or system error. See log file for details. • Invalid type ID • Name is already used by a reserved type. • Cannot add a type to the Alert Management category. • Calculation is invalid. Code: 401, Unauthorized • User token is invalid. • User lacks privileges. Code: 403, Forbidden - Contact system administrator.
<i>Notes</i>	User Defined Properties are customized in-application parameters and values for personal or shared use. User defined property types are grouped in categories. Use this call to create a new type under an existing category.

Chapter 19

Schema parameters

AlertRuntimeStatus

Integer (\$int32)

The alert status.

0 = Clear

1 = Warning

2 = Alarm

3 = Off

4 = OutOfService

AlertThresholdCreatePointRequestDto

projectPointId [integer(\$int32)]

The project point's unique ID number.

inputProjectPointType [ProjectPointTypeinteger(\$int32)]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

1 = InputSignal

2 = OutputClusterDistance

4 = OutputSignalPredictedUpper

5 = OutputSignalPredictedLower

6 = OutputSignalPredictedLinearRegression

7 = OutputSignalDeviationRelative

8 = OutputSignalDeviationAbsolute

9 = OutputSignalDeviationContribution

11 = OutputSignalActualValue

12 = OutputSignalPredictedValue

13 = OutputSignalAlarmState

alertThresholdType [AlertThresholdTypeinteger(\$int32)]

These are high or low alerts or warnings. Temporary alerts are a single-use threshold that can be set up while leaving existing thresholds alone. Out-of-range thresholds are deprecated.

1 = HighOutOfRange

2 = HighAlert

3 = HighWarning

4 = LowOutOfRange

5 = LowAlert

6 = LowWarning

7 = HighTemporaryAlert

8 = LowTemporaryAlert

thresholdValue [number(\$double)]

The numeric value of the threshold.

persistenceDenomination [AlertThresholdPersistenceDenominationinteger(\$int32)]

Persistence restricts when the point goes into alert. It either measures the ratio of how frequently (how often within a number of events) the value exceeds the threshold, or how long (the amount of time, in seconds) the value spends beyond the threshold. Absolute persistence is deprecated.

1 = PercentageOfTime

2 = PercentageOfCount

3 = Absolute

note [string]

maxLength: 50000

minLength: 0

This may describe why the values were chosen or why some thresholds are disabled.

minCount [integer(\$int32)]

This is the number of times (of the Out of total amount) the value must cross the threshold to trigger the alert.

secondsOrEventCount [integer(\$int32)]

This is the total number of seconds or events AVEVA Predictive Analytics observes to determine if the minimum threshold crossings have occurred.

isCritical [boolean]

This means the threshold is so important that the word "critical" appears in the Alerts list when the project point is in alert. This distinguishes it from its neighbors and calls more attention to the point.

AlertThresholdPersistenceDenomination

AlertThresholdPersistenceDenomination [integer(\$int32)]

Persistence restricts when the point goes into alert. It either measures the ratio of how frequently (how often within a number of events) the value exceeds the threshold, or how long (the amount of time, in seconds) the value spends beyond the threshold. Absolute persistence is deprecated.

1 = PercentageOfTime

2 = PercentageOfCount

3 = Absolute

AlertThresholdResponseDto

description

These are the response fields that the request returns.

alertThresholdId [integer(\$int32)]

The threshold's ID.

templateParent

AlertThresholdResponseDto

description

These are the response fields that the request returns.

alertThresholdId [integer(\$int32)]

The threshold's ID.

templateParent

alertThresholdType [AlertThresholdTypeinteger(\$int32)]

These are high or low alerts or warnings. Temporary alerts are a single-use threshold that can be set up while leaving existing thresholds alone. Out-of-range thresholds are deprecated.

1 = HighOutOfRange

2 = HighAlert

3 = HighWarning

4 = LowOutOfRange

5 = LowAlert

6 = LowWarning

7 = HighTemporaryAlert

8 = LowTemporaryAlert

projectPointId [integer(\$int32)]

The project point's ID.

projectPointType [ProjectPointTypeinteger(\$int32)]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

1 = InputSignal

2 = OutputClusterDistance

4 = OutputSignalPredictedUpper

5 = OutputSignalPredictedLower

6 = OutputSignalPredictedLinearRegression

7 = OutputSignalDeviationRelative

8 = OutputSignalDeviationAbsolute

9 = OutputSignalDeviationContribution

11 = OutputSignalActualValue

12 = OutputSignalPredictedValue

13 = OutputSignalAlarmState

projectPointUnits [string]

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

persistenceDenomination [AlertThresholdPersistenceDenominationinteger(\$int32)]

Persistence restricts when the point goes into alert. It either measures the ratio of how frequently (how often within a number of events) the value exceeds the threshold, or how long (the amount of time, in seconds) the value spends beyond the threshold. Absolute persistence is deprecated.

1 = PercentageOfTime

2 = PercentageOfCount

3 = Absolute

thresholdValue [number(\$double)]

The threshold's numeric value.

thresholdPercent [number(\$double)]

The amount of time for the threshold evaluation.

thresholdTimeSeconds [integer(\$int32)]

The amount of time the threshold is crossed before the alert is triggered.

thresholdCount [integer(\$int32)]

The number of events for the threshold evaluation.

thresholdMin [integer(\$int32)]

The number of threshold crossings required to trigger an alert.

alertNote [string]

nullable: true

An explanation of the threshold.

isActive [boolean]

Whether the threshold is enabled.

criticality [integer(\$int32)]

This means the threshold is so important that the word "critical" appears in the Alerts list when the project point is in alert. This distinguishes it from its neighbors and calls more attention to the point.

alertThresholdType [AlertThresholdTypeinteger(\$int32)]

These are high or low alerts or warnings. Temporary alerts are a single-use threshold that can be set up while leaving existing thresholds alone. Out-of-range thresholds are deprecated.

1 = HighOutOfRange

2 = HighAlert

3 = HighWarning

4 = LowOutOfRange

5 = LowAlert

6 = LowWarning

7 = HighTemporaryAlert

8 = LowTemporaryAlert

projectPointId [integer(\$int32)]

The project point's ID.

projectPointType [ProjectPointTypeinteger(\$int32)]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

1 = InputSignal

2 = OutputClusterDistance

4 = OutputSignalPredictedUpper

5 = OutputSignalPredictedLower

6 = OutputSignalPredictedLinearRegression

7 = OutputSignalDeviationRelative

8 = OutputSignalDeviationAbsolute

9 = OutputSignalDeviationContribution

11 = OutputSignalActualValue

12 = OutputSignalPredictedValue

13 = OutputSignalAlarmState

projectPointUnits [string]

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

persistenceDenomination [AlertThresholdPersistenceDenominationinteger(\$int32)]

Persistence restricts when the point goes into alert. It either measures the ratio of how frequently (how often within a number of events) the value exceeds the threshold, or how long (the amount of time, in seconds) the value spends beyond the threshold. Absolute persistence is deprecated.

1 = PercentageOfTime

2 = PercentageOfCount

3 = Absolute

thresholdValue [number(\$double)]

The threshold's numeric value.

thresholdPercent [number(\$double)]

The amount of time for the threshold evaluation.

thresholdTimeSeconds [integer(\$int32)]

The amount of time the threshold is crossed before the alert is triggered.

thresholdCount [integer(\$int32)]

The number of events for the threshold evaluation.

thresholdMin [integer(\$int32)]

The number of threshold crossings required to trigger an alert.

alertNote [string]

nullable: true

An explanation of the threshold.

isActive [boolean]

Whether the threshold is enabled.

criticality [integer(\$int32)]

This means the threshold is so important that the word "critical" appears in the Alerts list when the project point is in alert. This distinguishes it from its neighbors and calls more attention to the point.

AlertThresholdType

AlertThresholdType [integer(\$int32)]

These are high or low alerts or warnings. Temporary alerts are a single-use threshold that can be set up while leaving existing thresholds alone. Out-of-range thresholds are deprecated.

1 = HighOutOfRange

2 = HighAlert

3 = HighWarning

4 = LowOutOfRange

5 = LowAlert

6 = LowWarning

7 = HighTemporaryAlert

8 = LowTemporaryAlert

AlertThresholdUpdateRequestDto

thresholdId [integer(\$int32)]

The threshold ID number

isActive [boolean]

Whether the threshold is enabled

alertThresholdType [AlertThresholdTypeinteger(\$int32)]

These are high or low alerts or warnings. Temporary alerts are a single-use threshold that can be set up while leaving existing thresholds alone. Out-of-range thresholds are deprecated.

1 = HighOutOfRange

2 = HighAlert

3 = HighWarning

4 = LowOutOfRange

5 = LowAlert

6 = LowWarning

7 = HighTemporaryAlert

8 = LowTemporaryAlert

thresholdValue [number(\$double)]

The numeric value of the threshold.

persistenceDenomination [AlertThresholdPersistenceDenominationinteger(\$int32)]

Persistence restricts when the point goes into alert. It either measures the ratio of how frequently (how often within a number of events) the value exceeds the threshold, or how long (the amount of time, in seconds) the value spends beyond the threshold. Absolute persistence is deprecated.

1 = PercentageOfTime

2 = PercentageOfCount

3 = Absolute

note [string]

maxLength: 50000

minLength: 0

nullable: true

This may describe why the values were chosen or why some thresholds are disabled.

minCount [integer(\$int32)]

This is the number of times (of the Out of total amount) the value must cross the threshold to trigger the alert.

secondsOrEventCount [integer(\$int32)]

This is the total number of seconds or events AVEVA Predictive Analytics observes to determine if the minimum threshold crossings have occurred.

isCritical [boolean]

This means the threshold is so important that the word "critical" appears in the Alerts list when the project point is in alert. This distinguishes it from its neighbors and calls more attention to the point.

CalculationPointGetResponseDto

pointName [string]

nullable: true

The name of the calc point.

pointTypeMetricDescription [string]

nullable: true

The metric description, if applicable.

calculationPointType [CalculationPointTypeinteger(\$int32)]

Where the calculation point was created.

1 = ElapsedTime

2 = Project

3 = Template

description [string]

maxLength: 250

nullable: true

The point's description.

digitalGroupId [integer(\$int32)]

nullable: true

If applicable, the digital group.

extendedDescription [string]

maxLength: 250

nullable: true

The point's extended description.

extendedId [string]

maxLength: 250

nullable: true

id [integer(\$int32)]

The point's ID.

units [string]

maxLength: 50

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

CalculationPointType

CalculationPointType [integer(\$int32)]

Where the calculation point was created.

1 = ElapsedTime

2 = Project

3 = Template

CalculationPointUpdateRequestDto

description [string]

maxLength: 250

nullable: true

The point's description.

digitalGroupId [integer(\$int32)]

nullable: true

If applicable, the digital group.

extendedDescription [string]

maxLength: 250

nullable: true

The point's extended description.

extendedId [string]

maxLength: 250

nullable: true

id* integer(\$int32)

The point's ID.

units [string]

maxLength: 50

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

CalculationPointUpdateRequestDto

description [string]

maxLength: 250

nullable: true

The point's description.

digitalGroupId [integer(\$int32)]

nullable: true

If applicable, the digital group.

extendedDescription [string]

maxLength: 250

nullable: true

The point's extended description.

extendedId [string]

maxLength: 250

nullable: true

id [integer(\$int32)]

The point's ID.

units [string]

maxLength: 50

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

Coordinate

timeStamp [string(\$date-time)]**value [number(\$double)]**

DataRecordStatus

DataRecordStatus [integer(\$int32)]

The data quality.

0 = Good

1 = Offline

2 = Uninitialized

3 = Error

4 = Questionable

DeviationDirection

DeviationDirection [integer(\$int32)]

The direction the deviation takes, compared to the threshold, that determines whether the fault signature is a match.

0 = Normal

1 = Up

2 = UpDown

-1 = Down

DeviationFormat

DeviationFormat [integer(\$int32)]

The deviation format.

0 = Relative

1 = Absolute

DigitalGroupDto

id [integer(\$int32)]

The group's ID.

description [string]

nullable: true

The description of the group.

Fault Detail

faultId [integer(\$int32)]

The fault diagnostic's ID.

faultName [string]

The fault diagnostic's name.

activeAverage [number(\$double)]

The active average fault match.

fullAverage [number(\$double)]

The full average fault match.

isValid [boolean]

Whether the fault is properly mapped to a template.

FaultDiagnosticGetResponseDto

faultId [integer(\$int32)]

The fault's ID.

faultName [string]

The fault's name.

description [string]

The fault's behavioral description.

nextsteps [string]

Suggestions for how to begin remediating the fault.

faultDiagnosticProperties

userDefinedPropertyTypeId [integer(\$int32)]

The user defined property Types ID.

userDefinedPropertyId [integer(\$int32)]

The user defined properties' ID.

numericValue [number(\$double)]

If applicable, a customized number.

textValue [string]

If applicable, customized text.

FaultDiagnosticProperty

userDefinedPropertyTypeId [integer(\$int32)]

nullable: true

The user defined property Types ID.

userDefinedPropertyId [integer(\$int32)]

nullable: true

The user defined properties' ID.

numericValue [number(\$double)]

nullable: true

If applicable, a customized number.

textValue [string]

nullable: true

If applicable, customized text.

FaultDiagnosticsDetailsGetResponseDto

projectId [integer(\$int32)]

The project's ID.

projectName [string]

nullable: true

The project's name.

faultDetails

nullable: true

FaultDetail**faultId [integer(\$int32)]**

The fault diagnostic's ID.

faultName [string]

nullable: true

The fault diagnostic's name.

activeAverage [number(\$double)]

The active average fault match.

fullAverage [number(\$double)]

The full average fault match.

isValid [boolean]

Whether the fault is properly mapped to a template.

FaultDiagnosticsDetailsRequestDto

projectIds [integer(\$int32)]

maxItems: 5

The unique ID(s) of up to five project(s).

frequencySeconds [integer(\$int32)]

The sample frequency, in seconds.

startDate [string(\$date-time)]

The starting point of the observed time span, in UTC.

endDate [string(\$date-time)]

The ending point of the observed time span, in UTC.

FaultDiagnosticsGetSummaryResponseDto

activeAverage [number(\$double)]

The average fault match of active signatures.

fullAverage [number(\$double)]

The average fault match including full signature info.

activeRecent [number(\$double)]

This shows the match of only the latest active values (within the time span) for the fault.

fullRecent [number(\$double)]

This shows the match of only the latest full signature values (within the time span) for the fault.

faultMatches

nullable: true

FaultDiagnosticTimeSeriesDataPoint

fullValue [number(\$double)]

The point's value at a timestamp.

fullState [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

value [number(\$double)]

The point's value at a timestamp.

state [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

childState [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

timeStamp [string(\$date-time)]

The point's timestamp, in UTC.

inputPointSignatureSummaries

nullable: true

InputPointSignatureSummary

inputPointId [integer(\$int32)]

nullable: true

The point's ID.

upThresholdMin [number(\$double)]

nullable: true

The minimum threshold for rising deviation behavior.

upThresholdMax [number(\$double)]

nullable: true

The maximum threshold for rising deviation behavior.

downThresholdMin [number(\$double)]

nullable: true

The minimum threshold for falling deviation behavior.

downThresholdMax [number(\$double)]

nullable: true

The maximum threshold for falling deviation behavior.

Metric

id [integer(\$int32)]

The metric's ID.

name [string]

nullable: true

The metric's name.

deviationFormat [DeviationFormatinteger(\$int32)]

The deviation format.

0 = Relative

1 = Absolute

deviationDirection [DeviationDirectioninteger(\$int32)]

The direction the deviation takes, compared to the threshold, that determines whether the fault signature is a match.

0 = Normal

1 = Up

2 = UpDown

-1 = Down

average [number(\$double)]

This is the average match of the fault signature over the selected time span.

recent [number(\$double)]

This shows the match of only the latest values (within the time span) for the fault signature.

effectiveFullWeightPercentage [number(\$double)]

How much the fault signature contributes to the fault at the template level.

effectiveActiveWeightPercentage [number(\$double)]

How much the fault signature contributes to the fault at the project level.

faultMatches

nullable: true

FaultTimeSeriesDataPoint

value [number(\$double)]

The point's value at a timestamp.

state [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

childState [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

timeStamp [string(\$date-time)]

The point's timestamp, in UTC.

FaultDiagnosticState

FaultDiagnosticState [integer(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

FaultDiagnosticTimeSeriesDataPoint

fullValue [number(\$double)]

The point's value at a timestamp.

fullState [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

value [number(\$double)]

The point's value at a timestamp.

state [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

childState [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

timeStamp [string(\$date-time)]

The point's timestamp, in UTC.

FaultTimeSeriesDataPoint

value [number(\$double)]

The point's value at a timestamp.

state [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

childState [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

timeStamp [string(\$date-time)]

The point's timestamp, in UTC.

FilterStatus

FilterStatus [integer(\$int32)]

The status of the RTS.

0 = On

1 = Off

2 = OutOfService

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

ForecastData

predictions

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

upperLimit

ForecastCore

values

nullable: true

Coordinate

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

lowerLimit

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

recordCount [integer(\$int32)]

ForecastGetResponseDto

fittedValues

ForecastData

predictions

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

upperLimit

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

lowerLimit

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

forecastValues

ForecastData

predictions

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

upperLimit

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

lowerLimit

ForecastCore

values

nullable: true

Coordinate

timeStamp [string(\$date-time)]

value [number(\$double)]

start [string(\$date-time)]

end [string(\$date-time)]

recordCount [integer(\$int32)]

projectPointId [integer(\$int32)]

ForecastModelTypes

ForecastModelTypes [integer(\$int32)]

0 = LinearRegression

1 = Spline

ForecastRequestDto

frequency [string(\$date-span)]

startDate [string(\$date-time)]

values

nullable: true

[number (\$double)]

enableSmoothing [boolean]
noOfRecordsToPredict [integer(\$int32)]
modelType [ForecastModelTypesinteger(\$int32)]
 0 = LinearRegression
 1 = Spline
timeToFailureLimit [number(\$double)]
projectPointId [integer(\$int32)]

HistorianPointGetResponseDto

frequency [string(\$date-span)]
startDate [string(\$date-time)]
values
 nullable: true
 [number(\$double)]
enableSmoothing [boolean]
noOfRecordsToPredict [integer(\$int32)]
modelType [ForecastModelTypesinteger(\$int32)]
 0 = LinearRegression
 1 = Spline
timeToFailureLimit [number(\$double)]
projectPointId [integer(\$int32)]

HistorianPointUpdateRequestDto

pointName [string]
 nullable: true
 The point's name.
flatLineTimeMinutes [integer(\$int32)]
 maximum: 2147483647
 minimum: 0
 nullable: true
 How much time since the last change in the data value.

highOutOfRangeLimit [number(\$double)]

nullable: true

The threshold for high out of range (legacy points only).

lowOutOfRangeLimit [number(\$double)]

nullable: true

The threshold for low out of range (legacy points only).

manualOverride [boolean]

Whether quality pre-processing is ignored.

recoveryTimeMinutes integer(\$int32)

maximum: 2147483647

minimum: 0

nullable: true

How much time passes between connection attempts.

status [boolean]

nullable: true

Whether the point status (e.g., good, bad) from the RTS is imported with the data point.

description [string]

maxLength: 250

nullable: true

The point's description.

digitalGroupId [integer(\$int32)]

nullable: true

If applicable, the digital group.

extendedDescription [string]

maxLength: 250

nullable: true

The point's extended description.

extendedId [string]

maxLength: 250

nullable: true

id [integer(\$int32)]

The point's ID.

units [string]

maxLength: 50

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

HistoricalArchiveResponseDto

flatLineTimeMinutes [integer(\$int32)]

maximum: 2147483647

minimum: 0

nullable: true

How much time since the last change in the data value.

highOutOfRangeLimit [number(\$double)]

nullable: true

The threshold for high out of range (legacy points only).

lowOutOfRangeLimit [number(\$double)]

nullable: true

The threshold for low out of range (legacy points only).

manualOverride [boolean]

Whether quality pre-processing is ignored.

recoveryTimeMinutes [integer(\$int32)]

maximum: 2147483647

minimum: 0

nullable: true

How much time passes between connection attempts.

status [boolean]

nullable: true

Whether the point status (e.g., good, bad) from the RTS is imported with the data point.

description [string]

maxLength: 250

nullable: true

The point's description.

digitalGroupId [integer(\$int32)]

nullable: true

If applicable, the digital group.

extendedDescription [string]

maxLength: 250

nullable: true

The point's extended description.

extendedId [string]

maxLength: 250

nullable: true

id [integer(\$int32)]

The point's ID.

units [string]

maxLength: 50

nullable: true

The units (e.g., psi, degrees, rpm) of the point.

InputPointSignatureSummary

inputPointId [integer(\$int32)]

nullable: true

The point's ID.

upThresholdMin [number(\$double)]

nullable: true

The minimum threshold for rising deviation behavior.

upThresholdMax [number(\$double)]

nullable: true

The maximum threshold for rising deviation behavior.

downThresholdMin [number(\$double)]

nullable: true

The minimum threshold for falling deviation behavior.

downThresholdMax [number(\$double)]

nullable: true

The maximum threshold for falling deviation behavior.

metric

Metric

id [integer(\$int32)]

The metric's ID.

name [string]

nullable: true

The metric's name.

deviationFormat [DeviationFormatinteger(\$int32)]

The deviation format.

0 = Relative

1 = Absolute

deviationDirection [DeviationDirectioninteger(\$int32)]

The direction the deviation takes, compared to the threshold, that determines whether the fault signature is a match.

0 = Normal

1 = Up

2 = UpDown

-1 = Down

average [number(\$double)]

This is the average match of the fault signature over the selected time span.

recent [number(\$double)]

This shows the match of only the latest values (within the time span) for the fault signature.

effectiveFullWeightPercentage [number(\$double)]

How much the fault signature contributes to the fault at the template level.

effectiveActiveWeightPercentage [number(\$double)]

How much the fault signature contributes to the fault at the project level.

faultMatches

nullable: true

FaultTimeSeriesDataPoint

value [number(\$double)]

The point's value at a timestamp.

state [FaultDiagnosticStateinteger(\$int32)]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

childState [**FaultDiagnosticStateinteger(\$int32)**]

The value's relationship to a threshold.

0 = NoMatch

1 = FullMatch

2 = InterpolatedMatch

timeStamp [**string(\$date-time)**]

The point's timestamp, in UTC.

Metric

id [**integer(\$int32)**]

The metric's ID.

name [**string**]

nullable: true

The metric's name.

OutputPointInfo

inputPointId [**integer(\$int32)**]

outputPointId [**integer(\$int32)**]

outputPointType [**ProjectPointTypeinteger(\$int32)**]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

1 = InputSignal

2 = OutputClusterDistance

4 = OutputSignalPredictedUpper

5 = OutputSignalPredictedLower

6 = OutputSignalPredictedLinearRegression

7 = OutputSignalDeviationRelative

8 = OutputSignalDeviationAbsolute

9 = OutputSignalDeviationContribution

11 = OutputSignalActualValue

12 = OutputSignalPredictedValue

13 = OutputSignalAlarmState

outputPointRecordsData

nullable: true

OutputPointRecordData

timestamp [string(\$date-time)]

value [number(\$double)]

filterStatus [FilterStatusinteger(\$int32)]

The status of the RTS.

0 = On

1 = Off

2 = OutOfService

dataStatus [DataRecordStatusinteger(\$int32)]

The data quality.

0 = Good

1 = Offline

2 = Uninitialized

3 = Error

4 = Questionable

alertRuntimeStatus [AlertRuntimeStatusinteger(\$int32)]

The alert status.

0 = Clear

1 = Warning

2 = Alarm

3 = Off

4 = OutOfService

childDataStatus [DataRecordStatusinteger(\$int32)]

The data quality.

0 = Good

1 = Offline

2 = Uninitialized

3 = Error

4 = Questionable

childAlertRuntimeStatus [AlertRuntimeStatusinteger(\$int32)]

The alert status.

- 0 = Clear
- 1 = Warning
- 2 = Alarm
- 3 = Off
- 4 = OutOfService

OutputPointRecordData

timestamp [string(\$date-time)]

value [number(\$double)]

filterStatus [FilterStatusinteger(\$int32)]

The status of the RTS.

- 0 = On
- 1 = Off
- 2 = OutOfService

dataStatus [DataRecordStatusinteger(\$int32)]

The data quality.

- 0 = Good
- 1 = Offline
- 2 = Uninitialized
- 3 = Error
- 4 = Questionable

alertRuntimeStatus [AlertRuntimeStatusinteger(\$int32)]

The alert status.

- 0 = Clear
- 1 = Warning
- 2 = Alarm
- 3 = Off
- 4 = OutOfService

childDataStatus [DataRecordStatusinteger(\$int32)]

The data quality.

- 0 = Good

- 1 = Offline
- 2 = Uninitialized
- 3 = Error
- 4 = Questionable

childAlertRuntimeStatus [AlertRuntimeStatusinteger(\$int32)]

The alert status.

- 0 = Clear
- 1 = Warning
- 2 = Alarm
- 3 = Off
- 4 = OutOfService

OutputPointsArchiveStatusesDto

projectId [integer(\$int32)]

The project's ID.

archivedPointTypes

nullable: true

ProjectPointType [integer(\$int32)]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

- 1 = InputSignal
- 2 = OutputClusterDistance
- 4 = OutputSignalPredictedUpper
- 5 = OutputSignalPredictedLower
- 6 = OutputSignalPredictedLinearRegression
- 7 = OutputSignalDeviationRelative
- 8 = OutputSignalDeviationAbsolute
- 9 = OutputSignalDeviationContribution
- 11 = OutputSignalActualValue
- 12 = OutputSignalPredictedValue
- 13 = OutputSignalAlarmState

OutputPointsHistoricalArchiveRequestDto

frequencySeconds [integer(\$int32)]

startDate [string(\$date-time)]

endDate [string(\$date-time)]

inputPointIds [integer(\$int32)]

outputPointTypes [ProjectPointTypeinteger(\$int32)]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

1 = InputSignal

2 = OutputClusterDistance

4 = OutputSignalPredictedUpper

5 = OutputSignalPredictedLower

6 = OutputSignalPredictedLinearRegression

7 = OutputSignalDeviationRelative

8 = OutputSignalDeviationAbsolute

9 = OutputSignalDeviationContribution

11 = OutputSignalActualValue

12 = OutputSignalPredictedValue

13 = OutputSignalAlarmState

PointSelectorHierarchy

rtsDisplay [string]

nullable: true

sensorPoints

nullable: true

SensorPointDto

alertState [SystemPointAlertManagedStateinteger(\$int32)]

The condition (alert shape) of a historian point whose value has exceeded a threshold.

0 = ActionDue

1 = New

2 = AcknowledgeIncreased

3 = AcknowledgeNormal

4 = AcknowledgePending

5 = Clear

extendedDesc [string]

nullable: true

The historian point extended description.

extendedId [string]

nullable: true

The historian point extended name.

index [integer(\$int32)]

pointName [string]

nullable: true

The historian point name.

runtimeStatus [SystemPointRuntimeStatusinteger(\$int32)]

The current status of the historian point in relationship to the alert or warning threshold.

1 = InService

2 = OutOfService

systemPointDesc [string]

nullable: true

The description of the system point.

systemPointDisplay [string]

nullable: true

The points display name.

systemPointId [integer(\$int32)]

The system point's ID.

units [string]

nullable: true

The unit type for the historian point.

canClearAlert [boolean]

Whether the user has privileges to clear the alert.

assignedTo [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

assignedToDisplay [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

highOutOfRangeLimit [number(\$double)]

nullable: true

If applicable, the high value at which the preprocessor distinguishes bad quality data.

lowOutOfRangeLimit [number(\$double)]

nullable: true

If applicable, the low value at which the preprocessor distinguishes bad quality data.

ProjectFaultDiagnosticsGetResponseDto

faultId [integer(\$int32)]

The fault ID.

faultName [string]

nullable: true

The name of the fault.

description [string]

nullable: true

This is the fault diagnostic's behavioral description, as defined by your organization.

nextSteps [string]

nullable: true

These are suggestions for how to begin remediating the fault, as defined by your organization.

activeAverage [number(\$double)]

This is the average match of the active fault signatures over the selected time span.

fullAverage [number(\$double)]

This is the average match of the full fault signatures over the selected time span.

activeRecent [number(\$double)]

This shows the match of only the latest values (within the time span) for the active fault signatures.

fullRecent [number(\$double)]

This shows the match of only the latest values (within the time span) for the full fault signatures.

pointIds [integer(\$int32)]

nullable: true

The unique IDs of the project points.

ProjectPointType

ProjectPointType [integer(\$int32)]

This is a type of point or the alert threshold group for the threshold. Typically this is the actual value (11), absolute signal deviation (8), or relative signal deviation (7).

- 1 = InputSignal
- 2 = OutputClusterDistance
- 4 = OutputSignalPredictedUpper
- 5 = OutputSignalPredictedLower
- 6 = OutputSignalPredictedLinearRegression
- 7 = OutputSignalDeviationRelative
- 8 = OutputSignalDeviationAbsolute
- 9 = OutputSignalDeviationContribution
- 11 = OutputSignalActualValue
- 12 = OutputSignalPredictedValue
- 13 = OutputSignalAlarmState

SensorAlertStateDto

ids [integer(\$int32)]

alertManagedState

SystemPointAlertManagedState [integer(\$int32)]

The condition (alert shape) of a historian point whose value has exceeded a threshold.

- 0 = ActionDue
- 1 = New
- 2 = AcknowledgeIncreased
- 3 = AcknowledgeNormal
- 4 = AcknowledgePending
- 5 = Clear

deferDateUtc [string(\$date-time)]

nullable: true

The date that an response was deferred until when acknowledging the alert, in UTC.

assignedToUserId [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

updateAssignTo [boolean]

SensorFilter

SensorFilter [integer(\$int32)]

A filtered subset of sensor information.

710 = NoFilter

715 = InAlert

720 = NewAlert

725 = AcknowledgedAlerts

730 = ReturnedToService

735 = ManualOverride

SensorListDto

alertAgeDays [number(\$double)]

nullable: true

How many days the longest uncleared alert has been open (changed from clear to new).

alertDurationDays [number(\$double)]

nullable: true

How many cumulative days the historian point has been in alert.

alertPersistencePercentage [number(\$double)]

nullable: true

The Alert Duration divided by the Alert Age, as a percentage.

alertState [SystemPointAlertManagedStateinteger(\$int32)]

The condition (alert shape) of a historian point whose value has exceeded a threshold.

0 = ActionDue

1 = New

2 = AcknowledgeIncreased

3 = AcknowledgeNormal

4 = AcknowledgePending

5 = Clear

alertTotal [integer(\$int32)]

nullable: true

The cumulative number of alerts the historian point has had.

assignedTo [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

assignedToDisplay [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

canClearAlert [boolean]

Whether the user has permission to clear an alert.

checkForStatus [boolean]**earliestAlertDate [string(\$date-time)]**

nullable: true

When the historian point first went into alert, in UTC.

extendedDesc [string]

nullable: true

The historian point extended description.

extendedDescId [string]

nullable: true

The historian point extended name.

eventHistoryVisualization [string]

nullable: true

Based on the time span defined in the Event History Time Span setting, this bar chart shows how much of the time span (if any) the historian point has been in its most recent alert state.

flatLineTimeMinutes [integer(\$int32)]

nullable: true

How many minutes the data value has remained constant.

highThresholdLimit [number(\$double)]

nullable: true

If configured in the Client application, the high value at which the preprocessor distinguishes bad quality data.

impactedProjectsIds [integer(\$int32)]

nullable: true

Which projects use the historian point.

impactedProjectsCount [integer(\$int32)]

How many projects use the historian point.

latestAlertDate [string(\$date-time)]

nullable: true

When the historian point most recently went into alert.

lowThresholdLimit [number(\$double)]

nullable: true

If applicable, the low value at which the preprocessor distinguishes bad quality data.

recoveryTimeMinutes [integer(\$int32)]

nullable: true

How many minutes the preprocessor attempts to recover bad data.

rtsName [string]

nullable: true

The name of the historian point's real-time service.

runtimeStatus [SystemPointRuntimeStatusinteger(\$int32)]

The current status of the historian point in relationship to the alert or warning threshold.

1 = InService

2 = OutOfService

sensorIssueReason [SystemPointAlertStateReasoninteger(\$int32)]

Why the historian point values were removed from the data.

0 = Good

1 = ManualOverride

2 = Historian

3 = Oor

4 = Flatline

systemPointDesc [string]

nullable: true

The description of the system point.

systemPointId [integer(\$int32)]

The ID of the system point.

systemPointName [string]

nullable: true

The name of the system point.

units [string]

nullable: true

The unit type for the historian point.

SensorPointDto

alertState [SystemPointAlertManagedStateinteger(\$int32)]

The condition (alert shape) of a historian point whose value has exceeded a threshold.

0 = ActionDue

1 = New

2 = AcknowledgeIncreased

3 = AcknowledgeNormal

4 = AcknowledgePending

5 = Clear

extendedDesc [string]

nullable: true

The historian point extended description.

extendedId [string]

nullable: true

The historian point extended name.

index [integer(\$int32)]

pointName [string]

nullable: true

The historian point name.

runtimeStatus [SystemPointRuntimeStatusinteger(\$int32)]

The current status of the historian point in relationship to the alert or warning threshold.

1 = InService

2 = OutOfService

systemPointDesc [string]

nullable: true

The description of the system point.

systemPointDisplay [string]

nullable: true

The points display name.

systemPointId [integer(\$int32)]

The system point's ID.

units [string]

nullable: true

The unit type for the historian point.

canClearAlert [boolean]

Whether the user has privileges to clear the alert.

assignedTo [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

assignedToDisplay [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

highOutOfRangeLimit [number(\$double)]

nullable: true

If applicable, the high value at which the preprocessor distinguishes bad quality data.

lowOutOfRangeLimit [number(\$double)]

nullable: true

If applicable, the low value at which the preprocessor distinguishes bad quality data.

SensorPointSelectorDto

pointSelectors

nullable: true

PointSelectorHierarchy

rtsDisplay [string]

nullable: true

sensorPoints

nullable: true

SensorPointDto

alertState [SystemPointAlertManagedStateinteger(\$int32)]

The condition (alert shape) of a historian point whose value has exceeded a threshold.

0 = ActionDue

1 = New

2 = AcknowledgeIncreased

3 = AcknowledgeNormal

4 = AcknowledgePending

5 = Clear

extendedDesc [string]

nullable: true

The historian point extended description.

extendedId [string]

nullable: true

The historian point extended name.

index [integer(\$int32)]

pointName [string]

nullable: true

The historian point name.

runtimeStatus [SystemPointRuntimeStatusinteger(\$int32)]

The current status of the historian point in relationship to the alert or warning threshold.

1 = InService

2 = OutOfService

systemPointDesc [string]

nullable: true

The description of the system point.

systemPointDisplay [string]

nullable: true

The points display name.

systemPointId [integer(\$int32)]

The system point's ID.

units [string]

nullable: true

The unit type for the historian point.

canClearAlert [boolean]

Whether the user has privileges to clear the alert.

assignedTo [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

assignedToDisplay [string]

nullable: true

The user to whom the alert (if applicable) was assigned.

highOutOfRangeLimit [number(\$double)]

nullable: true

If applicable, the high value at which the preprocessor distinguishes bad quality data.

lowOutOfRangeLimit [number(\$double)]

nullable: true

If applicable, the low value at which the preprocessor distinguishes bad quality data.

SensorRequestDto

assetIds [integer(\$int32)]

The list of asset IDs.

templateIds [integer(\$int32)]

The list of template IDs.

listIds [integer(\$int32)]

The list of list IDs.

resultsFormat [SensorResultsFormatinteger(\$int32)]

The format for sensor information.

1 = ListView

2 = SummaryView

filters

nullable: true

SensorFilter [integer(\$int32)]

A filtered subset of sensor information.

710 = NoFilter

715 = InAlert

720 = NewAlert

725 = AcknowledgedAlerts

730 = ReturnedToService

735 = ManualOverride

SensorResultsFormat

SensorResultsFormat [integer(\$int32)]

The format for sensor information.

- 1 = ListView
- 2 = SummaryView

SensorResultsType

SensorResultsType [integer(\$int32)]

The organization of sensor information.

- 1 = Asset
- 2 = Template
- 3 = UserList

SystemPointAlertManagedState

SystemPointAlertManagedState [integer(\$int32)]

The condition (alert shape) of a historian point whose value has exceeded a threshold.

- 0 = ActionDue
- 1 = New
- 2 = AcknowledgeIncreased
- 3 = AcknowledgeNormal
- 4 = AcknowledgePending
- 5 = Clear

SystemPointAlertStateReason

SystemPointAlertStateReason [integer(\$int32)]

Why the historian point values were removed from the data.

- 0 = Good
- 1 = ManualOverride
- 2 = Historian
- 3 = Oor
- 4 = Flatline

SystemPointRuntimeStatus

SystemPointRuntimeStatus [integer(\$int32)]

The current status of the historian point in relationship to the alert or warning threshold.

1 = InService

2 = OutOfService

UserDefinedPropertyCreateRequestDto

userDefinedPropertyTypeId [integer(\$int32)]

The type's ID.

systemClassification

UserDefinedPropertySystemClassification [integer(\$int32)]

Classification for the case management category.

0 = UserDefined

1 = Open

2 = Closed

id [integer(\$int32)]

The detail's ID.

name [string]

maxLength: 100

The detail's name.

isEnabled [boolean]

The detail's visibility in the drop-down menu on its category screen or feature.

displayOrder [integer(\$int32)]

For a list configuration, the detail's order in the list.

numericValue [number(\$double)]

nullable: true

For a list configuration, the numeric value.

textValue [string]

maxLength: 1000

nullable: true

For a list configuration, the text value.

UserDefinedPropertyGetResponseDto

userDefinedPropertyTypeId [integer(\$int32)]

The user define property type's ID.

systemClassification [integer(\$int32)]

0 is for one or more user defined details; 1 is Open; 2 is Closed.

createdBy [string]

nullable: true

If applicable, the automatically generated name of the user who created the detail.

createdDate [string(\$date-time)]

nullable: true

If applicable, the automatically generated date when the detail was created, in UTC.

modifiedBy [string]

nullable: true

If applicable, the automatically generated name of the most recent user to change the detail.

modifiedDate [string(\$date-time)]

nullable: true

If applicable, the automatically generated date of most recent change, in UTC.

id [integer(\$int32)]

The detail's ID.

name [string]

maxLength: 100

The detail's name.

isEnabled [boolean]

The detail's visibility in the drop-down menu on its category screen or feature.

displayOrder [integer(\$int32)]

For a list configuration, the detail's order in the list.

numericValue [number(\$double)]

nullable: true

For a list configuration, the numeric value.

textValue [string]

maxLength: 1000

nullable: true

For a list configuration, the text value.

UserDefinedPropertyRemapRequestDto

userDefinedPropertyId [integer(\$int32)]

The original detail's ID.

remapToUserDefinedPropertyId [integer(\$int32)]

The replacement detail's ID.

UserDefinedPropertySystemClassification

UserDefinedPropertySystemClassification [integer(\$int32)]

Classification for the case management category.

0 = UserDefined

1 = Open

2 = Closed

UserDefinedPropertyTypeCategory

UserDefinedPropertyTypeCategory [integer(\$int32)]

The categories for user defined property types.

1 = AlertManagement

2 = CaseManagement

3 = Asset

4 = Project

5 = FaultDiagnostic

UserDefinedPropertyTypeConfig

UserDefinedPropertyTypeConfig [integer(\$int32)]

The configuration options for a type.

0 = List

1 = Text

2 = Numeric

3 = Calculated

UserDefinedPropertyTypeCreateRequestDto

systemClassification [UserDefinedPropertyTypeSystemClassificationinteger(\$int32)]

Classification of reserved types, and user defined (0).

0 = UserDefined

1 = AlarmClearType

2 = AlarmActionType

3 = AlarmClassificationType

4 = CaseCategoryType

5 = FaultConditionType

6 = ResponseType

7 = ImpactType

8 = CaseState

9 = AvoidedCosts

categoryId [UserDefinedPropertyTypeCategoryinteger(\$int32)]

The categories for user defined property types.

1 = AlertManagement

2 = CaseManagement

3 = Asset

4 = Project

5 = FaultDiagnostic

name [string]

maxLength: 100

The type's name.

description [string]

nullable: true

The type's description.

isEnabled [boolean]

Whether the type is enabled.

calcEquation [string]

nullable: true

If applicable, the type's calculation.

typeConfig [UserDefinedPropertyTypeConfiginteger(\$int32)]

The configuration options for a type.

- 0 = List
- 1 = Text
- 2 = Numeric
- 3 = Calculated

UserDefinedPropertyTypeGetResponseDto

id [integer(\$int32)]

The type's ID.

systemClassification [UserDefinedPropertyTypeSystemClassificationinteger(\$int32)]

Classification of reserved types, and user defined (0).

- 0 = UserDefined
- 1 = AlarmClearType
- 2 = AlarmActionType
- 3 = AlarmClassificationType
- 4 = CaseCategoryType
- 5 = FaultConditionType
- 6 = ResponseType
- 7 = ImpactType
- 8 = CaseState
- 9 = AvoidedCosts

categoryId [UserDefinedPropertyTypeCategoryinteger(\$int32)]

The categories for user defined property types.

- 1 = AlertManagement
- 2 = CaseManagement
- 3 = Asset
- 4 = Project
- 5 = FaultDiagnostic

usageCount [integer(\$int32)]

The number of records that use this type.

createdBy [string]

nullable: true

The user who created the type.

createdDate [string(\$date-time)]

nullable: true

The date when the type was created, in UTC.

modifiedBy [string]

nullable: true

The user who most recently changed the type.

modifiedDate [string(\$date-time)]

nullable: true

The date of most recent change, in UTC.

name [string]

maxLength: 100

The type's name.

description [string]

nullable: true

The type's description.

isEnabled [boolean]

Whether the type is enabled.

calcEquation [string]

nullable: true

If applicable, the type's calculation.

typeConfig [UserDefinedPropertyTypeConfiginteger(\$int32)]

The configuration options for a type.

0 = List

1 = Text

2 = Numeric

3 = Calculated

UserDefinedPropertyTypeSystemClassification

UserDefinedPropertyTypeSystemClassification [integer(\$int32)]

Classification of reserved types, and user defined (0).

0 = UserDefined

1 = AlarmClearType

2 = AlarmActionType

- 3 = AlarmClassificationType
- 4 = CaseCategoryType
- 5 = FaultConditionType
- 6 = ResponseType
- 7 = ImpactType
- 8 = CaseState
- 9 = AvoidedCosts

UserDefinedPropertyTypeUpdateRequestDto

id [integer(\$int32)]

The type's ID.

name [string]

maxLength: 100

The type's name.

description [string]

nullable: true

The type's description.

isEnabled [boolean]

Whether the type is enabled.

calcEquation [string]

nullable: true

If applicable, the type's calculation.

typeConfig [UserDefinedPropertyTypeConfiginteger(\$int32)]

The configuration options for a type.

0 = List

1 = Text

2 = Numeric

3 = Calculated

UserDefinedPropertyUpdateRequestDto

id [integer(\$int32)]

The detail's ID.

name [string]

maxLength: 100

The detail's name.

isEnabled [boolean]

The detail's visibility in the drop-down menu on its category screen or feature.

displayOrder [integer(\$int32)]

For a list configuration, the detail's order in the list.

numericValue [number(\$double)]

nullable: true

For a list configuration, the numeric value.

textValue [string]

maxLength: 1000

nullable: true

For a list configuration, the text value.